

SGA 3, Exposé IX
Groups of Multiplicative Type:
Homomorphisms into a Group Scheme
English Reconstruction

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Exposé IX

Groups of Multiplicative Type: Homomorphisms into a Group Scheme

by A. Grothendieck⁰

1. Definitions

Definition 1.1. — *Let S be a prescheme and let G be a group prescheme over S . The group G is said to be of multiplicative type if it is locally diagonalizable for the faithfully flat quasi-compact topology (cf. IV 6.3); that is, if for every $s \in S$, there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \rightarrow U$ such that*

$$G' = G \times_S S'$$

is a diagonalizable S' -group (I 4.4 and VIII 1.1).

The group G is said to be of quasi-isotrivial multiplicative type if it is locally diagonalizable even for the étale topology (IV 6.3); that is, if in the preceding definition one can take $S' \rightarrow U$ to be étale and surjective. Equivalently—as one sees by taking the disjoint union of the schemes S' attached to the various $s \in S$ —there is an étale surjective morphism $S' \rightarrow S$ such that $G' = G \times_S S'$ is a locally diagonalizable S' -group.

If one can moreover choose $S' \rightarrow S$ to be finite, étale, and surjective, then G is said to be of isotrivial multiplicative type.

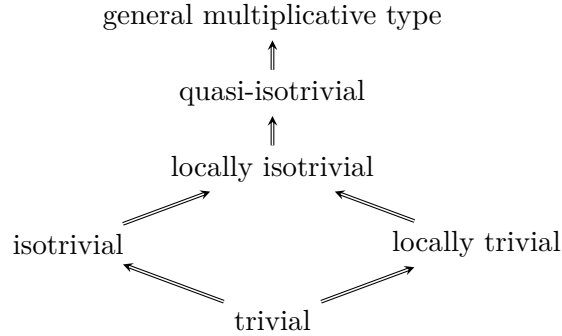
Finally, G is said to be a group of locally trivial multiplicative type (respectively, of locally isotrivial multiplicative type) if every $s \in S$ has an open neighborhood U such that $G \times_S U$ is a diagonalizable U -group (respectively, is of isotrivial multiplicative type; in other words, there is a finite étale surjective morphism $S' \rightarrow U$ such that $G \times_S S'$ is a diagonalizable S' -group).

Comments 1.2. — The five preceding notions are all obtained from the notion of a diagonalizable group by localization, with respect to five different “topologies” associated with (Sch).

It is understood in general that, when the word “locally” is not qualified by naming the topology under consideration, the Zariski topology is meant, in accordance with established terminology. In the terminology introduced here, “of locally trivial multiplicative

⁰Editor’s note: version 1.0 of November 8, 2009. Additions were made in the proof of 3.6 bis, in 4.4–7, 5.0–6, 6.1, 7.1, and 8.2. Sections 8.1 and 4.5 remain to be revised.

type” is equivalent to “locally diagonalizable” in VIII 1.1; likewise, “trivial” translates as “diagonalizable.” The five notions introduced above fit into the following implication diagram, which follows from the corresponding relations among the topologies involved:



For practical purposes, let us point out at once that every group of multiplicative type that we shall encounter will be quasi-isotrivial. Thus we shall see in the next exposé (X 4.5) that, when G is of finite type over S , it is automatically quasi-isotrivial; we shall nevertheless give examples in which it is not locally isotrivial. We shall likewise see that G may be locally trivial without being isotrivial, and hence *a fortiori* without being trivial. This readily implies that the inclusions in the preceding diagram are strict.

On the other hand, we shall also see there (X 5.16) that, when S is locally noetherian and normal—or, more generally, geometrically unibranch—every group of multiplicative type and finite type over S is necessarily isotrivial; moreover, it is trivial as soon as it is locally trivial (or even merely trivial over a dense open subset). This explains why most groups of multiplicative type encountered in practice will doubtless be isotrivial, all the more so because we shall later see that the maximal tori of semisimple group schemes are automatically isotrivial.

Definition 1.3. — *Let S and G be as in 1.1. The group G is called a torus if, for the faithfully flat quasi-compact topology, it is locally isomorphic to a group of the form $\mathbf{G}_m^r$, where r is an integer with $r \geq 0$.*

With the notation of 1.1, this says that S' may be chosen so that G' is isomorphic to a group of the form $(\mathbf{G}_{m,S'})^r$. The integer r depends on $s \in S$: it is the dimension of the fiber

$$G_s = G \otimes_S \kappa(s).$$

It is a locally constant function of s , as is seen at once. This observation should be generalized as follows.

Definition 1.4. — *Let G be a diagonalizable group scheme over a field k . Thus G is isomorphic to a group $D_k(M)$, where M is an ordinary commutative group determined up to isomorphism by this condition—more precisely,*

$$M \simeq \text{Hom}_{k\text{-gr}}(G, \mathbf{G}_{m,k})$$

(VIII 1.3). *The isomorphism class of M is called the type of the diagonalizable group G ; it is plainly invariant under extension of the base field.*

Now let G be a group scheme of multiplicative type over an arbitrary prescheme S . For every $s \in S$, there is an extension k of $\kappa(s)$ such that $G \times_S \text{Spec}(k)$ is a diagonalizable k -group.¹ Its type is independent of the chosen extension by the preceding observation and

¹Editor’s note: indeed, by hypothesis there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \rightarrow U$ such that $G_{S'}$ is diagonalizable. For every $s' \in S'$ above s , the field $\kappa(s')$ is an extension of $\kappa(s)$, and $G_{s'} = G \times_S \text{Spec}(\kappa(s'))$ is diagonalizable.

is called the type of G at s , or the type of G_s . In particular, if G itself is diagonalizable, and hence isomorphic to a group of the form $D_S(M)$, then for every $s \in S$ the type of G at s is the isomorphism class of M .

More generally, if G is a group of multiplicative type over S and M is an ordinary commutative group, then G is said to be of type M if it is of type M at every point $s \in S$; equivalently, if G is locally isomorphic to $D_S(M)$ for the faithfully flat quasi-compact topology.

Remark 1.4.1. ² a) For a group G of multiplicative type over S , the function

$$s \longmapsto \text{the type of } G_s$$

is locally constant on S . Indeed, with the notation of 1.1, if G' is of type M , then G is of type M over the neighborhood U of s . Consequently S has a canonical decomposition as a disjoint union of preschemes S_i such that, for every i , G_{S_i} is of type M_i , where the commutative groups M_i are pairwise nonisomorphic.

b) In particular, if S is connected, the type of the fibers of G is constant; that is, there is a commutative group M such that G is of type M . Finally, if G is a torus, the type of G at s is determined by the integer $\dim G_s$: indeed, G_s is of type $\mathbb{Z}^n$, where $n = \dim G_s$.

Remark 1.5. — a) It follows immediately from Definitions 1.1 and 1.3 that these notions are “compatible with base extension.” Thus, if G is a group prescheme over S and $S' \rightarrow S$ is a base-change morphism, then whenever G is of multiplicative type (respectively, of isotrivial multiplicative type, and so on), the same is true of G' .

b) If, moreover, $S' \rightarrow S$ is faithfully flat and quasi-compact, then G' being of multiplicative type, respectively a torus, implies the same for G . If $S' \rightarrow S$ is also étale (respectively, finite étale), then G' being quasi-isotrivial (respectively, isotrivial) implies the same for G .

c) Finally, return to an arbitrary base-change morphism $f : S' \rightarrow S$. If $s' \in S'$ and $s = f(s')$, then the type of G' at s' is the same as that of G at s , since

$$G'_{s'} = G_s \otimes_{\kappa(s)} \kappa(s').$$

2. Extension of Certain Properties of Diagonalizable Groups to Groups of Multiplicative Type

The extensions in question are essentially trivial consequences of the results of the preceding Exposé, in view of the definitions in 1.1 and the “local” nature (for the faithfully flat and quasi-compact topology) of the results under consideration.

Definition 2.0. ³ We denote by $\mathcal{M}$ the set of faithfully flat and quasi-compact morphisms.

Proposition 2.1. Let G be a group of multiplicative type over a prescheme S . Then:

- (a) G is faithfully flat over S , and affine over S (a fortiori, quasi-compact over S);
- (b) G is of finite type over S if and only if G is of finite presentation over S , if and only if, for every $s \in S$, the type of G at s is given by a commutative group of finite type;

²Editor’s note: the number 1.4.1 has been added to the remarks that follow, for use in later references.

³Editor’s note: This definition has been added; it occurs in Propositions 2.3 and 2.7.

- (c) G is finite over S if and only if, for every $s \in S$, the type of G at s is given by a finite commutative group, if and only if (when S is quasi-compact) G is of finite type over S and is annihilated by an integer $n > 0$;
- (c') G is integral over S if and only if, for every $s \in S$, the type of G at s is given by a torsion commutative group;
- (d) G is the unit group if and only if, for every $s \in S$, the type of G at s is given by the zero commutative group;
- (e) G is smooth over S if and only if, for every $s \in S$, the type of G at s is given by a commutative group of finite type whose torsion subgroup has order prime to the characteristic of $\kappa(s)$.

These statements follow from VIII 2.1, taking into account that the properties in question descend under faithfully flat and quasi-compact morphisms (cf. SGA 1, VIII, or EGA IV₂, §2).

Using VIII 3.5, one likewise obtains:

Proposition 2.2. *Let G be a group of multiplicative type and of finite type over S . For every integer $n \neq 0$, the kernel ${}_nG = \ker(n \cdot \text{id}_G)$ is a group of multiplicative type, finite over S .*

Proposition 2.3. *Let G be a group of multiplicative type over the prescheme S , acting freely on the right on an S -prescheme X that is affine over S . Then:*

- (a) *the equivalence relation defined by G in X is $\mathcal{M}$ -effective (IV 3.4), and $Y = X/G$ is affine over S ;*
- (b) *if, moreover, X is of finite presentation (respectively, of finite type) over S , then the same is true of Y .*

The first assertion follows from VIII 5.1, which treats the case in which G is diagonalizable, and from IV 3.5.2, which allows one to reduce to that case. Indeed, faithfully flat and quasi-compact morphisms $S' \rightarrow S$ are effective descent morphisms for the fibered category of affine schemes over schemes: if Y' is affine over S' and is equipped with a descent datum relative to $S' \rightarrow S$, then this datum is effective, that is, Y' comes from a scheme Y affine over S (cf. SGA 1, VIII 2.1).

For the second assertion, one is likewise reduced to the diagonalizable case of VIII 5.8, since the finiteness conditions under consideration descend under faithfully flat and quasi-compact morphisms (SGA 1, VIII 3.3 and 3.6⁴). Proceeding as in VIII, Corollaries 5.5–5.7, one deduces from 2.3:

Corollary 2.4. *Under the hypotheses of 2.3, the graph morphism*

$$X \times_S G \longrightarrow X \times_S X$$

is a closed immersion. For every section σ of X over S , the corresponding morphism

$$G \longrightarrow X, \quad g \longmapsto \sigma \cdot g,$$

is a closed immersion.

In particular:

⁴Editor's note: See also V, 9.1, or EGA IV₂, 2.7.1.

Corollary 2.5. *Let $u : G \rightarrow H$ be a monomorphism of S -preschemes in groups, with G of multiplicative type and H affine over S . Then u is a closed immersion, $H/G = Y$ exists and is affine over S , and H is a principal homogeneous bundle over Y under G_Y .*

Remark 2.6. Let $u : G \rightarrow H$ be a monomorphism of S -preschemes in groups, where G is of multiplicative type and of finite type over S , and H is of finite presentation and separated over S . One can then show that u is a closed immersion by using VIII 7.12 (see Remark VIII 7.13(b)).

Proposition 2.7. *Let $u : G \rightarrow H$ be a homomorphism of S -groups of multiplicative type, with H of finite type over S . Then:*

- (i) $G_0 = \ker u$ is an S -group of multiplicative type; the equivalence relation defined by G_0 in G is $\mathcal{M}$ -effective, and hence (IV 3.4) u factors as

$$G \longrightarrow G/G_0 = I \longrightarrow H,$$

where $I \rightarrow H$ is a monomorphism⁵ of S -groups. The group I is of multiplicative type, the equivalence relation in H defined by I is $\mathcal{M}$ -effective, and consequently $H_0 = H/I$ exists and is also of multiplicative type;

- (ii) H_0 and I are of finite type, and so is G_0 when G is.

Proof. Proceeding as for 2.3, one is reduced to the case in which G and H are diagonalizable; then 2.7 reduces to VIII 3.4.

We record the following corollaries.

Corollary 2.8.

- (a) *Let S be a prescheme. The category of S -groups of multiplicative type and of finite type is an abelian category.*
- (b) *Let $u : G \rightarrow H$ be a homomorphism in this category. It is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in this category if and only if u is a monomorphism of preschemes (respectively, u is faithfully flat, respectively, u is an isomorphism of preschemes).*

It is enough to note that the product of two S -groups of multiplicative type is again of multiplicative type (which is immediate), and is plainly of finite type over S if both factors are. The rest of 2.8 follows immediately from 2.7; the details are left to the reader.

Corollary 2.9. *Let $u : G \rightarrow H$ be a homomorphism of S -groups of multiplicative type and of finite type. Let U be the set of points $s \in S$ for which $u_s : G_s \rightarrow H_s$ is a monomorphism (respectively, faithfully flat, respectively, an isomorphism). Then U is both open and closed, and the induced homomorphism $u_U : G_U \rightarrow H_U$ is a monomorphism (respectively, faithfully flat, respectively, an isomorphism).*

Let P (respectively, Q) be the kernel (respectively, cokernel) of u . By 2.7, Q exists and P and Q are of multiplicative type; moreover, formation of P and Q commutes with every base change $S' \rightarrow S$, in particular with the formation of fibers. On the other hand, u is a monomorphism (respectively, faithfully flat, respectively, an isomorphism) if and only if $P = 0$ (respectively, $Q = 0$, respectively, $P = Q = 0$). These remarks reduce us to checking the following assertion: if R is an S -group of multiplicative type, then the set U

⁵Editor's note: Indeed, even a closed immersion.

of points $s \in S$ for which $R_s = 0$ is open and closed, and $R_U = 0$. This is contained in Remark 1.4.1(a).

Corollary 2.10. *Let G be an S -group of multiplicative type and of finite type, and let H and H' be two subgroups of multiplicative type.*

- (a) $H'' = H \cap H'$ is a subgroup of G of multiplicative type.
- (b) Let U be the set of $s \in S$ for which $H_s \subseteq H'_s$ (respectively, $H_s = H'_s$). Then U is open and closed, and $H_U \subseteq H'_U$ (respectively, $H_U = H'_U$).

Here, of course, the intersection sign in $H \cap H'$ denotes the intersection in the functorial sense, namely $H \times_G H'$, which is plainly an S -subgroup of G . Likewise, the inclusion sign (respectively, the equality sign) denotes the order relation (respectively, equality) between subfunctors of H , not inclusion (respectively, equality) of the underlying sets.

Applying 2.7 first to the inclusion morphism $H \rightarrow G$, one finds that H is of finite type; the same holds for H' . It then follows from 2.8 that $H \cap H'$ is of multiplicative type and of finite type. (Note that the canonical functor from the category considered in 2.8 to the category **(Sch)**/ S commutes with finite inverse limits.)

Formation of $H \cap H'$ commutes with base extension, in particular with formation of the fibers. Moreover, $H \subseteq H'$ (respectively, $H = H'$) is equivalent to $H'' = H$ (respectively, to $H'' = H$ and $H'' = H'$). Consider the canonical homomorphisms $H'' \rightarrow H$ and $H'' \rightarrow H'$. Then U is the set of points $s \in S$ for which the induced homomorphism $H''_s \rightarrow H_s$ is an isomorphism (respectively, for which $H''_s \rightarrow H_s$ and $H''_s \rightarrow H'_s$ are isomorphisms).

By 2.9, it follows that U is open and closed, and that $H''_U \rightarrow H_U$ (respectively, $H''_U \rightarrow H_U$ and $H''_U \rightarrow H'_U$) are isomorphisms. This gives the desired conclusion.

Proposition 2.11. *Let S be a prescheme, let G be an S -group of multiplicative type and of finite type over S , let H be a subgroup of G of multiplicative type, and put $K = G/H$, a quotient group of G of multiplicative type.*

- (i) Suppose G is trivial, that is, diagonalizable. There is a partition of S into open and closed subsets S_i such that, for every i , H_{S_i} and K_{S_i} are diagonalizable. In particular, if S is connected, H and K are diagonalizable.
- (ii) The same assertion holds with “diagonalizable” replaced by “isotrivial,” provided that S is connected, locally connected, or quasi-compact.
- (iii) If G is locally trivial (respectively, locally isotrivial, respectively, quasi-isotrivial), then the same is true of K and H .

Proof. (i) By hypothesis, $G = D_S(M)$, where M is a commutative group of finite type. For each quotient group M_i of M , let $H_i = D_S(M_i)$ be the corresponding diagonalizable subgroup of G (VIII 3.1). Let S_i be the set of points $s \in S$ for which $H_s = (H_i)_s$. By 2.10, S_i is open and closed, and $H_{S_i} = (H_i)_{S_i}$; thus H_{S_i} is diagonalizable, and hence so is K_{S_i} (VIII 3.1). The S_i are plainly pairwise disjoint, and we claim that they cover S . Indeed, for every s , the group H_s , being a subgroup of the diagonalizable group G_s , is diagonalizable (cf. 8.1 below⁶). Hence H_s has the form $D_{\kappa(s)}(M_i)$, by VIII 1.5 and 3.2(b). Restricting to the family of nonempty S_i , assertion (i) follows.

(ii) By hypothesis, there exists a finite surjective étale morphism $S' \rightarrow S$ such that $G_{S'}$ is diagonalizable. Every point of S' therefore has an open and closed neighborhood U'

⁶Editor’s note: The erroneous reference to IV 4.7.5 has been corrected. Note that Section 8 of the present Exposé is independent of Sections 3–7.

such that $H_{U'}$ and $K_{U'}$ are diagonalizable. The image U of U' in S is open and closed, and $S' \rightarrow S$ still induces a finite surjective étale morphism $U' \rightarrow U$. Thus every point $s \in S$ has an open and closed neighborhood U such that H_U and K_U are isotrivial. Assertion (ii) follows at once.

(iii) This follows at once from (i), (ii), and the definitions. Moreover, the quasi-isotrivial case will also follow from the more general fact “finite type $\implies$ quasi-isotrivial,” announced in 1.2 (cf. X 4.5).

3. Infinitesimal Properties: Lifting and Conjugation Theorems

The fundamental infinitesimal properties of groups of multiplicative type follow from the following theorem.

Theorem 3.1. *Let S be an affine scheme, let H be an S -group of multiplicative type, and let $\mathcal{F}$ be a quasi-coherent sheaf on S on which H acts (I 4.7). Then*

$$H^i(H, \mathcal{F}) = 0 \quad \text{for } i > 0,$$

where H^i denotes the Hochschild cohomology studied in Exposé I, §5.

Indeed, by loc. cit., 5.3, Hochschild cohomology is expressed, in terms of the affine rings A and B of S and G , and the A -module M defining $\mathcal{F}$, as the cohomology of a complex of A -modules $C^\bullet(H, M)$, whose formation commutes with every base change $A \rightarrow A'$. Consequently, for a base change $A \rightarrow A'$ with A' flat over A , one has

$$H^i(H', \mathcal{F}') = H^i(H, \mathcal{F}) \otimes_A A'.$$

If A' is moreover faithfully flat over A , it is therefore enough, in order to prove that $H^i(H, \mathcal{F})$ vanishes, to prove that $H^i(H', \mathcal{F}')$ vanishes. This observation reduces us to checking 3.1 in the case where G is diagonalizable; in that case it was proved in I 5.3.3.

Using the results of Exposé III, we shall derive from 3.1 several consequences of a geometric nature.

Theorem 3.2. *Let S be an affine scheme, let S_0 be an affine subscheme defined by an ideal J , let H be a group of multiplicative type over S , and let G be an arbitrary group prescheme over S . Let $u, v : H \rightarrow G$ be two group homomorphisms, and let $u_0, v_0 : H_0 \rightarrow G_0$ be the homomorphisms obtained from u, v by the base change $S_0 \rightarrow S$. If $J^2 = 0$ and $u_0 = v_0$, then there exists $g \in G(S)$ such that*

$$v = \text{int}(g) \circ u \quad \text{and} \quad g_0 = e.$$

This follows from III 2.1(ii) and 3.1 (the vanishing of H^1).

Corollary 3.3. *Under the preliminary hypotheses of 3.2, suppose in addition that G is smooth over S , but assume only that J is nilpotent (not necessarily square-zero). If there exists $g_0 \in G_0(S_0)$ such that $v_0 = \text{int}(g_0) \circ u_0$, then g_0 lifts to an element $g \in G(S)$ such that $v = \text{int}(g) \circ u$.*

By induction on an integer n such that $J^n = 0$, one is reduced to the case in which J is square-zero. Since G is smooth over S , one may lift g_0 to an element g' of $G(S)$. Replacing v by $v' = \text{int}(g') \circ u$, one is then reduced to the situation of 3.2.

Corollary 3.4. *Under the preliminary hypotheses of 3.2, with J nilpotent, suppose in addition that u is central (for example, that G is commutative). Then $u_0 = v_0$ implies $u = v$.*

Indeed, the reduction to the case in which J is square-zero is again immediate, and then it is enough to apply 3.2. In particular:

Corollary 3.5. *Let S, S_0, H, G be as in 3.2, with J nilpotent. Let $u : H \rightarrow G$ be a homomorphism of S -groups such that $u_0 : H_0 \rightarrow G_0$ is the unit homomorphism. Then u is the unit homomorphism.*

Theorem 3.6. *Let S be an affine scheme, let S_0 be a subscheme defined by a nilpotent ideal J , let H be a group of multiplicative type over S , let G be a group prescheme smooth over S , and let $u_0 : H_0 \rightarrow G_0$ be a homomorphism of the S_0 -groups obtained by base change $S_0 \rightarrow S$. Then there exists a homomorphism $u : H \rightarrow G$ of S -groups that lifts u_0 . Moreover, by 3.3, any two such liftings u, u' are conjugate by an element of $G(S)$ whose reduction in $G(S_0)$ is the identity element.*

This follows from III 2.1 and 2.3 and from 3.1: the vanishing of H^2 gives the existence of the lifting of u_0 , and the vanishing of H^1 gives uniqueness up to conjugation.

One may prove in the same way the following variants of 3.2–3.6. In fact, these variants imply the preceding statements when applied to the graph subgroups of the homomorphisms considered there.

Theorem 3.2 bis. *Let S be an affine scheme, let S_0 be a subscheme defined by an ideal J , let G be an S -group prescheme, and let H, H' be subgroup preschemes of multiplicative type. Let G_0, H_0, H'_0 be the groups over S_0 obtained by base change $S_0 \rightarrow S$. If $J^2 = 0$ and $H_0 = H'_0$, then there exists $g \in G(S)$ such that*

$$H' = \text{int}(g)(H) \quad \text{and} \quad g_0 = e.$$

Corollary 3.3 bis. *Under the preliminary hypotheses of 3.2 bis, suppose in addition that G is smooth over S , and that J is nilpotent (not necessarily square-zero). If $g_0 \in G_0(S_0)$ satisfies $H'_0 = \text{int}(g_0)(H_0)$, then g_0 lifts to an element $g \in G(S)$ such that $H' = \text{int}(g)(H)$.*

Corollary 3.4 bis. *Under the preliminary hypotheses of 3.2 bis, suppose that J is nilpotent and that H is central (for example, that G is commutative). Then $H_0 = H'_0$ implies $H = H'$.*

Theorem 3.6 bis. *Let S be an affine scheme, let S_0 be a subscheme defined by a nilpotent ideal J , let G be an S -group prescheme smooth over S , and let H_0 be a subgroup prescheme of $G_0 = G \times_S S_0$, of multiplicative type. Then:*

- (a) *there exists a subgroup prescheme H of G , flat over S , such that $H \times_S S_0 = H_0$;*
- (b) *such an H is necessarily of multiplicative type;*
- (c) *finally, by 3.2 bis, any two such liftings H, H' of H_0 are conjugate by an element $g \in G(S)$ whose reduction in $G(S_0)$ is the identity element.*

Let us indicate how one can prove 3.2 bis (of which 3.3 bis and 3.4 bis are immediate consequences) and assertion (a) of 3.6 bis. The latter follows from 3.1 (the vanishing of H^2) and III 4.37(ii). Similarly, 3.2 bis follows from 3.1 (the vanishing of H^1) and III 4.37(i), at least when G and H are smooth over S .

Using the results of the following Exposé, however, one can very simply deduce 3.2 bis and 3.6 bis, in the general form stated here, from 3.2 and 3.6, respectively. For 3.2 bis, it is enough to note that, by X 2.1, H and H' are isomorphic, which reduces us to 3.2.

For 3.6 bis, note that H_0 is necessarily of finite type over S_0 :⁷ indeed, as a subgroup prescheme of G_0 , which is smooth over S_0 , the group H_0 is locally of finite type over S_0 . But H_0 is affine over S_0 by 2.1(a), and is therefore of finite type over S_0 . By X 4.5, H_0 is consequently quasi-isotrivial, and hence comes, by X 2.1, from a group H of multiplicative type over S .⁸ By 3.6, there exists a homomorphism of S -groups $u : H \rightarrow G$ lifting the immersion $H_0 \hookrightarrow G_0$. Since H and H_0 (respectively, G and G_0) have the same underlying topological space, and since, for every $h \in H$, the morphism

$$\mathcal{O}_{G,u(h)} \longrightarrow \mathcal{O}_{H,h}$$

is surjective (because it is so after reduction modulo the nilpotent ideal J), the morphism u is also an immersion (cf. EGA I, 4.2.2).

Finally, every lifting H of H_0 to a flat subgroup of G is necessarily of multiplicative type by X 2.3,⁹ which also proves assertion (b) of 3.6 bis. The reader may verify that the results 3.2 bis–3.6 bis are not used in Exposé X, so there is no vicious circle.

Proposition 3.8. ¹⁰ *Let S be a prescheme, let S_0 be a closed subscheme defined by a locally nilpotent ideal J , let G be an S -group of multiplicative type, and let X be an S -prescheme locally of finite presentation. Suppose that G acts on X in such a way that $G_0 = G \times_S S_0$ acts trivially on $X_0 = X \times_S S_0$. Then G acts trivially on X .*

The proof is the proof of III 2.4(b). One may also reduce to loc. cit. by observing that 3.8 immediately reduces to the case in which G is diagonalizable, which is the case treated there.

Corollary 3.9. *Let S, S_0 be as above, and let $u : G \rightarrow H$ be a homomorphism of S -groups, with G of multiplicative type and H locally of finite presentation over S . Suppose that the homomorphism $u_0 : G_0 \rightarrow H_0$ obtained by base change $S_0 \rightarrow S$ is central. Then u is central.*

Indeed, it is enough to apply 3.8 with $X = H$, letting G act on H by

$$(g, h) \longmapsto \text{int}(u(g)) \cdot h = u(g)h u(g)^{-1}.$$

4. The Density Theorem

This theorem (cf. 4.7 below)*, together with the theorem¹¹ of Section 7, will be the essential tool in the present exposé and the two following ones for passing from the infinitesimal properties of groups of multiplicative type developed above to their “finite” properties.

Definition 4.1. *Let X be a prescheme. A family $(Z_i)_{i \in I}$ of subpreschemes of X is said to be schematically dense if, for every open subset U of X and every closed subprescheme U_0 of U containing all the $Z_i \cap U$, one has $U_0 = U$.*

A subprescheme Z of X is said to be schematically dense in X if the one-member family consisting of Z is schematically dense.

⁷Editor’s note: The original said, “by 2.1(b), since its fibers are.” The editors did not understand this argument and have replaced it by the argument that follows.

⁸Editor’s note: The sentence that follows has been expanded.

⁹Editor’s note: X 2.3 depends essentially on Theorem 3.6 of the present Exposé.

¹⁰Editor’s note: The numbering of the original has been retained; there is no 3.7.

*All the results from 4.1 through 4.6 are contained in EGA IV₃, 11.10, to which we refer the reader for a systematic account of the notion of schematic density.

¹¹Editor’s note: the “algebraicity” theorem 7.1.

It is immediate (cf. EGA IV₃, 11.10.1) that the definition is equivalent to saying that, for every open subset U of X , every section f of $\mathcal{O}_U$ that vanishes on each $Z_i \cap U$ is zero. Equivalently, the intersection of the kernels of the canonical homomorphisms

$$u_i : \mathcal{O}_X \longrightarrow (v_i)_*(\mathcal{O}_{Z_i})$$

is zero, where $v_i : Z_i \rightarrow X$ is the canonical immersion.¹² When X lies over a prescheme S , this is also equivalent to the following assertion: for every open subset U of X , every S -prescheme Y separated over S , and every pair of morphisms $u, v : U \rightarrow Y$ that coincide on the $Z_i \cap U$, one has $u = v$. Indeed, $u = v$ is equivalent to $U_0 = U$, where U_0 is the inverse image of the diagonal in $Y \times_S Y$ under the S -morphism $U \rightarrow Y \times_S Y$ defined by (u, v) . The diagonal is a closed subscheme of $Y \times_S Y$, so U_0 is a closed subscheme of U containing all the $Z_i \cap U$. Thus schematic density gives $U_0 = U$, hence $u = v$. Conversely, take $Y = \text{Spec } \mathcal{O}_S[T]$.

In the terminology introduced at the end of EGA I, 9.5, saying that the subscheme Z is schematically dense in X also means that X is the *closure subscheme* of Z in X .

Lemma 4.2. *Let X be a flat S -prescheme, and let $(Z_i)_{i \in I}$ be a family of subschemes of X , flat over S . Let S_0 be a subscheme of S defined by a nilpotent ideal $\mathcal{J}$, and suppose that the modules $\mathcal{J}^n / \mathcal{J}^{n+1}$ are locally free over S_0 . Let X_0, Z_{i0} be obtained from X, Z_i by the base change $S_0 \rightarrow S$. If $(Z_{i0})_{i \in I}$ is schematically dense in X_0 , then $(Z_i)_{i \in I}$ is schematically dense in X .*

Proof. Suppose $\mathcal{J}^{n+1} = 0$, where $n \geq 0$, and argue by induction on n ; the assertion is trivial for $n = 0$. Define S_m, X_m, Z_{im} by reduction modulo $\mathcal{J}^{m+1}$, as usual. The induction hypothesis already implies that $(Z_{i,n-1})_{i \in I}$ is schematically dense in X_{n-1} . After replacing X by an induced open subset, it is enough to prove that every section f of $\mathcal{O}_X$ that vanishes on all the Z_i is zero.

The section f_{n-1} of

$$\mathcal{O}_{X_{n-1}} = \mathcal{O}_X / \mathcal{J}^n \mathcal{O}_X$$

defined by f vanishes on the $Z_{i,n-1}$, hence is zero. Consequently, f is a section of $\mathcal{J}^n \mathcal{O}_X$. Since X is flat over S , there is an isomorphism

$$\mathcal{J}^n \mathcal{O}_X \xleftarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{X_0}, \quad \mathcal{E} = \mathcal{J}^n = \mathcal{J}^n / \mathcal{J}^{n+1}.$$

Similarly, since each Z_i is flat over S , the restriction f_i of f to Z_i may be regarded as a section of

$$\mathcal{J}^n \mathcal{O}_{Z_i} \xleftarrow{\sim} \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Z_{i0}}.$$

By hypothesis, all the f_i are zero. The module $\mathcal{E}$ is locally free by assumption, and therefore so is

$$\mathcal{F} = \mathcal{E} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{X_0}.$$

Thus f is a section of the locally free module $\mathcal{F}$ on X_0 whose restriction to every Z_{i0} is zero. Since the family $(Z_{i0})_{i \in I}$ is schematically dense in X_0 , it follows at once that $f = 0$. $\square$

Corollary 4.3. *Let X be a locally Noetherian S -prescheme, flat over S , and let $(Z_i)_{i \in I}$ be a family of subschemes of X , flat over S . Suppose that, for every $s \in S$, the family of fibers $(Z_{is})_{i \in I}$ is schematically dense in X_s . Then $(Z_i)_{i \in I}$ is schematically dense in X .*

Proof. After replacing X by an induced open subset, it is enough to prove that every section f of $\mathcal{O}_X$ whose restriction to all the Z_i is zero is itself zero.¹³ For this, it suffices to

¹²Editor's note: all the results on schematic density stated in EGA IV₃, §11.10 follow from the results on "separating families of homomorphisms" proved in *loc. cit.*, §11.9, to which some of the editor's notes below will refer.

¹³Editor's note: the original has been expanded in what follows.

show that the image of f in $\mathcal{O}_{X,x}$ is zero for every $x \in X$. Let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{X,x}$, and let $\mathfrak{m}_s$ be that of $\mathcal{O}_{S,s}$, where s is the image of x in S . Since $\mathcal{O}_{X,x}$ is Noetherian,

$$\bigcap_{n \in \mathbb{N}} \mathfrak{m}_x^n = 0, \quad \text{and hence, a fortiori,} \quad \bigcap_{n \in \mathbb{N}} \mathcal{O}_{X,x} \mathfrak{m}_s^n = 0.$$

It is therefore enough to show, for every integer n , that the section induced by f on

$$X \times_S \operatorname{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}_s^{n+1})$$

is zero. By hypothesis,

$$Z_{is} = Z_i \otimes_{\mathcal{O}_{S,s}} \kappa(s)$$

is a schematically dense family in

$$X_s = X \otimes_{\mathcal{O}_{S,s}} \kappa(s).$$

We are thus reduced to the case in which S is the spectrum of a local ring whose maximal ideal $\mathcal{J}$ is nilpotent, $S_0 = \operatorname{Spec} \kappa(s)$, and the hypotheses of Lemma 4.2 hold. The conclusion follows. $\square$

Lemma 4.4. *Let S be a locally Noetherian prescheme, let X be a locally Noetherian S -prescheme, flat over S , and let $(Z_i)_{i \in I}$ be a family of subpreschemes of X , flat over S . Suppose that, for every $s \in S$, the family of fibers $(Z_{is})_{i \in I}$ is schematically dense in X_s . Then, for every base change $S' \rightarrow S$, with S' not necessarily locally Noetherian, the family $(Z'_i)_{i \in I}$ is schematically dense in X' . In other words, $(Z_i)_{i \in I}$ is universally schematically dense in X relative to S (cf. EGA IV₃, Definition 11.10.8).*

*Proof.*¹⁴ Let f' be a section of $\mathcal{O}_{X'}$ over an open subset U' of X' , vanishing on the Z'_i ; we must show that $f' = 0$. Fix $s' \in S'$. It is enough to prove that f' vanishes at every point of U' above s' . If s is the image of s' in S , then, after replacing S, S' by the spectra of their local rings at s, s' , we may suppose that S, S' are local and that s, s' are their closed points. We may also suppose that X is affine, so that

$$\begin{aligned} S &= \operatorname{Spec}(A), & S' &= \operatorname{Spec}(A'), \\ X &= \operatorname{Spec}(B), & X' &= \operatorname{Spec}(B'), & B' &= B \otimes_A A', \end{aligned}$$

where A, A' are local, $A \rightarrow A'$ is a local homomorphism, and A, B are Noetherian. We may further suppose that U' is affine, of the form $X'_{g'}$, with $g' \in B'$.

For every A -subalgebra A'' of A' , put

$$S'' = \operatorname{Spec}(A'')$$

and let X'', Z''_i be obtained from X, Z_i by the base change $S'' \rightarrow S$. We obtain the diagram

$$\begin{array}{ccccc} Z^i & \longleftarrow & Z''_n & \longleftarrow & Z'^{i'} \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & X'' & \longleftarrow & X' \\ \downarrow & & \downarrow & & \downarrow \\ S & \longleftarrow & S'' & \longleftarrow & S'. \end{array}$$

¹⁴Editor's note: for another proof, reducing to the case in which S' is locally Noetherian and using 4.3 and 4.5 as here, see EGA IV₃, 11.9.16 and 11.9.12. In the last line of the proof of 11.9.16, replace 11.9.5 by 11.9.12.

In what follows, we restrict to those A'' that are localizations of A -subalgebras A'_1 of A' , of finite type over A , at $\mathfrak{m}' \cap A'_1$, where $\mathfrak{m}'$ is the maximal ideal of A' . Thus the homomorphisms $A \rightarrow A'' \rightarrow A'$ are local. These A'' form an increasing filtered family of subalgebras whose union, and hence whose direct limit, is A' . Likewise,

$$B' = \varinjlim_{A''} (B \otimes_A A'').$$

There therefore exist such an A'' and an element $g'' \in B'' = B \otimes_A A''$ whose image in B' is g' .

By Lemma 4.5 below, the property that $(Z_{is})_{i \in I}$ be schematically dense in X_s is preserved by every base change $S'' \rightarrow S$.¹⁵ Consequently, because each S'' is locally Noetherian, replacing S by a suitable S'' and X, Z_i by X'', Z''_i preserves the hypotheses of Lemma 4.4.

We may therefore suppose that g' comes from an element $g \in B$. After replacing X by the open subset $X_g = U$, we may suppose that $U' = X'$. In the same way, we may suppose that f' comes from a section f of $\mathcal{O}_X$ over X .

Let $Y = V(f)$ be the subscheme of X defined by f , and put

$$Y_i = Z_i \times_X Y.$$

Thus Y_i is the closed subscheme $V(f_i)$ of Z_i , where f_i is the section of $\mathcal{O}_{Z_i}$ induced by f . Denoting by Y', Y'_i the S' -schemes obtained by base change, we have

$$Y' = V(f'), \quad Y'_i = Z'_i \times_{X'} Y' = V(f'_i),$$

and analogous relations hold for Y'', Y''_i . The assertion that f' vanishes on the Z'_i is equivalent to

$$Y'_i = Z'_i \quad \text{for every } i.$$

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Let $\mathfrak{m}$ be the maximal ideal of A . For every $n \geq 0$, introduce the closed subscheme

$$S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$$

of S , and let X_n, Y_n, Z_n^i, Y_n^i be obtained from X, Y, Z_i, Y_i by the base change $S_n \rightarrow S$. More generally, for every S -prescheme P , write $P_n = P \times_S S_n$.

For every n and $i \in I$, consider the functor

$$F_n^i = \prod_{Z_n^i/S_n} Y_n^i/Z_n^i : (\mathbf{Sch})_{/S_n}^\circ \longrightarrow \mathbf{Set},$$

defined, for every P over S_n , by

$$F_n^i(P) = \Gamma((Y_n^i)_P/(Z_n^i)_P) = \begin{cases} \emptyset, & (Y_n^i)_P \neq (Z_n^i)_P, \\ \{\text{id}\}, & (Y_n^i)_P = (Z_n^i)_P. \end{cases}$$

Since S_n is local Artinian and Z_n^i is flat, hence essentially free, over S_n , VIII, 6.4 shows that each F_n^i is represented by a closed subprescheme T_n^i of S_n .

¹⁵Editor's note: the preceding sentence has been added, and the original has been expanded in what follows.

¹⁶Editor's note: the original added "at least at every point x' of X' above s' ," but this restriction seems unnecessary.

The completion $\widehat{A}$ of the local ring A is Noetherian because A is, and it is the projective limit of the rings A_n .¹⁷ Let K_n^i be the ideal of

$$A_n = A/\mathfrak{m}^{n+1}$$

defining T_n^i , and let $K^i = \varprojlim_n K_n^i$; this is an ideal of $\widehat{A}$.

For fixed i , integers $m \geq n$, and any S_n -prescheme P ,

$$(Y_n^i)_P = Y_i \times_S S_n \times_{S_n} P = Y_i \times_S P = (Y_m^i)_P,$$

and similarly $(Z_n^i)_P = (Z_m^i)_P$. It follows that F_n^i is the restriction $F_m^i \times_{S_m} S_n$ of F_m^i to $(\mathbf{Sch})_{/S_n}$, and therefore

$$T_n^i = T_m^i \times_{S_m} S_n.$$

We thus have a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_m^i & \longrightarrow & A_m & \longrightarrow & \mathcal{O}(T_m^i) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & K_n^i & \longrightarrow & A_n & \longrightarrow & \mathcal{O}(T_n^i) \longrightarrow 0. \end{array}$$

All the vertical arrows are surjective. It follows, first, that the projective system $(K_n^i)_{n \in \mathbb{N}}$ satisfies the Mittag-Leffler condition, so that

$$\varprojlim_n \mathcal{O}(T_n^i) \simeq \widehat{A}/K^i$$

as topological rings (cf. EGA 0_{III}, §13.2). Second, the map $K^i \rightarrow K_n^i$ is surjective (cf. [BEns], III, §7.4, Proposition 5). Hence

$$K_n^i \simeq \frac{K^i + \mathfrak{m}^{n+1}\widehat{A}}{\mathfrak{m}^{n+1}\widehat{A}}, \quad \mathcal{O}(T_n^i) \simeq (\widehat{A}/K^i) \otimes_A (A/\mathfrak{m}^{n+1}).$$

In other words, $(T_n^i)_n$ is an inductive system of affine Artinian schemes, and its inductive limit

$$T^i = \varinjlim_n T_n^i$$

is a closed formal subscheme of

$$\widehat{S} = \mathrm{Spf}(\widehat{A})$$

(cf. EGA I, §10), whose reduction modulo $\mathfrak{m}^{n+1}$ is T_n^i .

Let T be the closed formal subscheme of $\widehat{S}$ given by the intersection of the T^i , that is, defined by

$$K = \sum_i K^i.$$

Since $\widehat{A}$ is Noetherian, there is a finite subset J of I such that

$$K = \sum_{i \in J} K^i, \quad \text{equivalently} \quad T = \bigcap_{i \in J} T^i.$$

¹⁷Editor's note: the original has been expanded and corrected to show that the projective limit of the rings $\mathcal{O}(T_n^i)$ defines a closed formal subscheme of $\mathrm{Spf}(\widehat{A})$.

For every n , therefore,

$$K_n = \sum_{i \in J} K_n^i,$$

where K_n is the image of K in A_n .

Recall that f_i is the image of $f \in B = \mathcal{O}(X)$ in $\mathcal{O}(Z_i)$, and that f'_i is its image in

$$\mathcal{O}(Z'_i) = \mathcal{O}(Z_i) \otimes_A A'.$$

The relation $Y'_i = Z'_i$ is equivalent to $f'_i = 0$. Now $\mathcal{O}(Z_i) \otimes_A A'$ is the direct limit of the subalgebras $\mathcal{O}(Z_i) \otimes_A A''$, for A'' as above. Consequently, there is an A'' for which $Y''_i = Z''_i$. A priori A'' depends on i , but, because J is finite, one A'' can be chosen that works for every $i \in J$. Put

$$S'' = \text{Spec}(A''), \quad S''_{(n)} = S'' \times_S S_n.$$

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For every $i \in J$, the equality

$$Y''_i \times_{S''} S''_{(n)} = (Y''_i)_{S''_{(n)}} = Z''_i \times_{S''} S''_{(n)} = (Z''_i)_{S''_{(n)}}$$

shows, by the definition of T_n^i , that $S''_{(n)} \rightarrow S_n$ factors through T_n^i . It therefore factors through

$$T_n = \bigcap_{i \in J} T_n^i,$$

and hence through T_n^i for every $i \in I$. Let f'' be the image of f in $B'' = B \otimes_A A''$, and let $f''_{(n)}$ be its image in

$$B''_{(n)} = B'' / \mathfrak{m}^{n+1} B''.$$

The preceding factorization means that, for every n , the section $f''_{(n)}$ vanishes on all the $Z_i \times_S S''_{(n)}$.

Because $A \rightarrow A'' \rightarrow A'$ are local homomorphisms, the image s'' of s' in S'' is the closed point of S'' , corresponding to the maximal ideal $\mathfrak{m}''$ of A'' , and its image in S is s . *[The source prints B'' here; since $s'' \in S'' = \text{Spec}(A'')$, the intended ring is A'' .]* Fix n , and now let S''_n denote the n -th infinitesimal neighborhood of s'' in S'' :

$$S''_n = \text{Spec}(A'' / \mathfrak{m}''^{n+1}).$$

Put

$$B''_n = B'' / \mathfrak{m}''^{n+1} B'', \quad Z''_n = Z_i \times_S S''_n.$$

[The source appends $Z''_n = \text{Spec}(B''_n)$; the displayed equality would identify each Z''_n with X''_n , so the base-change definition is retained here.] Then the image f''_n of $f''_{(n)}$ in B''_n vanishes on every Z''_n . By Lemma 4.5 below, the family of fibers

$$Z''_0 = Z''_n \times_{S''_n} \kappa(s'') = Z_{is} \times_{\kappa(s)} \kappa(s'')$$

is schematically dense in

$$X'''_0 = X''_n \times_{S''_n} \kappa(s'') = X_s \times_{\kappa(s)} \kappa(s'').$$

¹⁸Editor's note: the original has been expanded and corrected, distinguishing $S''_{(n)}$ from the subscheme $S'' = S''(s'')$ introduced below.

Thus $S''_n, X''_n, (Z''_n)_{i \in I}$, and

$$S'''_0 = \operatorname{Spec} \kappa(s'')$$

satisfy the hypotheses of Lemma 4.2. It follows that $f''_n = 0$, that is,

$$f'' \in \mathfrak{m}''^{n+1} B'' \quad \text{for every } n.$$

Since B is Noetherian and $B'' = B \otimes_A A''$ is a localization of a finite-type B -algebra, B'' is Noetherian. Its local rings are therefore separated for their usual topology. Hence f'' is zero at every point x'' of X'' for which $\mathfrak{m}'' B''$ is contained in the maximal ideal of $\mathcal{O}_{X'', x''}$, that is, at every point of X'' above s'' . A fortiori, f' is zero at every point of X' above s' . $\square$

Lemma 4.5.0.¹⁹ *Let X be an S -prescheme, let $(Z_i)_{i \in I}$ be a family of subpreschemes of X , and let $S' \rightarrow S$ be faithfully flat. If $(Z'_i)_{i \in I}$ is schematically dense in X' , then $(Z_i)_{i \in I}$ is schematically dense in X .*

This follows from EGA IV₃, 11.10.5(i) and 11.9.10(i). It remains to give the proof of the following lemma.

Lemma 4.5. *Let X be a locally Noetherian prescheme²⁰ over a field k , let $(Z_i)_{i \in I}$ be a family of subpreschemes of X , let k'/k be a field extension, and let X', Z'_i be obtained from X, Z_i by base change. Then $(Z_i)_{i \in I}$ is schematically dense in X if and only if $(Z'_i)_{i \in I}$ is schematically dense in X' .*

Proof. The “if” direction follows from Lemma 4.5.0; let us prove the converse.²¹ First, we may suppose $X = \operatorname{Spec}(B)$ affine. For every i , let $(Z_{ij})_{j \in J_i}$ be an affine open covering of Z_i . Replacing $(Z_i)_{i \in I}$ by $(Z_{ij})_{(i,j) \in J}$, where

$$J = \coprod_{i \in I} J_i,$$

we may also suppose that every Z_i is affine.

Let

$$g \in B' = B \otimes_k k', \quad t' \in B'_g$$

be a section of $\mathcal{O}_{X'}$ on the affine open $U' = X'_g$ whose restriction to each $Z'_i \cap U'$ is zero. Thus its image in every

$$(\mathcal{O}(Z_i) \otimes_k k')_g$$

is zero. There exists a finite-type k -subalgebra A of k' such that

$$g \in B_A = B \otimes_k A, \quad t' \in (B_A)_g.$$

The map

$$\mathcal{O}(Z_i) \otimes_k A \longrightarrow \mathcal{O}(Z_i) \otimes_k k'$$

is injective, and therefore so is

$$(\mathcal{O}(Z_i) \otimes_k A)_g \longrightarrow (\mathcal{O}(Z_i) \otimes_k k')_g.$$

¹⁹Editor’s note: this lemma has been inserted here because it is used in the proofs of 4.5 and 4.7.

²⁰Editor’s note: this hypothesis is in fact superfluous; cf. EGA IV₃, 11.10.6 and 11.9.13.

²¹Editor’s note: the original has been expanded in what follows.

The hypothesis consequently implies that the image of t' in every $(\mathcal{O}(Z_i) \otimes_k A)_g$ is zero. It remains to show that $(Z_{iA})_{i \in I}$ is schematically dense in X_A , which follows from EGA IV₃, 11.9.10(b).²² $\square$

Let us record in passing the following result, which will be used in a later exposé.²³

Corollary 4.6. *Let X be a flat S -prescheme and U an open subset of X . Suppose that U_s is schematically dense in X_s for every $s \in S$, and that either X is locally Noetherian or X is locally of finite presentation over S . Then U is schematically dense in X .*

Proof. For simplicity, suppose U is retrocompact in X (cf. EGA IV₃, 11.10.10 and 11.9.17 for the general case). The locally Noetherian case is included only for reference; it is contained in 4.3.

In the second case, we may plainly suppose S and X affine. Then X and U are of finite presentation over $S = \operatorname{Spec}(A)$. The ring A is the direct limit of its finite-type $\mathbb{Z}$ -subalgebras A_i . The “patented procedure” already used (cf. EGA IV₃, 8.8.2 and 8.10.5(iii)) shows that there are an index i , an affine scheme X_i over

$$S_i = \operatorname{Spec}(A_i),$$

and an open subset U_i of X_i , from which X, U are obtained by the base change $S \rightarrow S_i$. Let E_i be the subset of S_i consisting of those $s \in S_i$ for which $(U_i)_s$ is schematically dense in $(X_i)_s$.

By EGA IV₂, 5.9.9 and 5.10.2, E_i is the set of points $s \in S_i$ for which $(U_i)_s$ contains the set

$$\operatorname{Ass} \mathcal{O}_{(X_i)_s}$$

of points associated with the structure sheaf of $(X_i)_s$. By EGA IV₃, 11.9.17.1, this is a constructible condition; hence E_i is constructible.²⁴

By 4.5, the inverse image of E_i under $S_j \rightarrow S_i$ is E_j , and its inverse image under $S \rightarrow S_i$ is the set E of those $s \in S$ for which U_s is schematically dense in X_s .²⁴ By hypothesis, $E = S$, which is also the inverse image of S_i under every $S \rightarrow S_i$. EGA IV₃, 8.3.11 therefore implies that there is a $j \geq i$ such that $E_j = S_j$; that is, $(U_j)_s$ is schematically dense in $(X_j)_s$ for every $s \in S_j$.

Apply 4.4 to (S_j, X_j, U_j) and the base change $S \rightarrow S_j$. It follows that U is schematically dense in X . Notice that here 4.4 is used only for a family with a finite index set (in fact, a singleton); in this case its proof simplifies considerably. $\square$

Theorem 4.7. *Let S be a prescheme, and let G be a group of multiplicative type and of finite type over S . For every integer $n > 0$, let ${}_nG$ be the kernel of $n \cdot \operatorname{id}_G$. Then the family of subpreschemes $({}_nG)_{n > 0}$ is schematically dense in G in the sense of Definition 4.1.*

Proof. We distinguish two cases.

(a) *S locally Noetherian.* By 4.3, we are reduced to the case in which S is the spectrum of a field k . By 4.5.0, we may suppose that k is algebraically closed and that G is diagonalizable, say

$$G = D_k(M),$$

where M is a commutative group of finite type. Then

$$M \simeq \Gamma \times \mathbb{Z}^r$$

²²Editor’s note: the original continued: “Using 4.3 for the $X_{A'}$, where A' is a quotient of A by a power of a maximal ideal, and Hilbert’s Nullstellensatz, one is reduced to the case where k' is a finite extension of k .” This reduction could be expanded and that argument pursued, since the case of a finite extension k'/k is somewhat simpler than the more general case treated in EGA IV₃, 11.9.10(b).

²³Editor’s note: this reference remains to be specified.

²⁴Editor’s note: the original has been expanded in this paragraph and the next.

with Γ finite, and consequently

$$G \simeq G_1 \times G_2, \quad G_1 = D(\Gamma), \quad G_2 = \mathbf{G}_m^r.$$

For n sufficiently divisible—namely, for n a multiple of the order of Γ —one has

$${}_nG = G_1 \times {}_nG_2, \quad \text{since} \quad {}_nG_1 = G_1.$$

Applying 4.3 again to the projection $G \rightarrow G_1$, we are reduced to G_2 , hence to the case $G = \mathbf{G}_m^r$. Since G is then reduced, schematic density of $({}_nG)_{n>0}$ is equivalent to density of $\bigcup_n {}_nG$ in the ordinary topology. As

$${}_nG = ({}_n\mathbf{G}_m)^r,$$

we are reduced to $G = \mathbf{G}_m$, an irreducible group of dimension one. The assertion now follows because $\bigcup_n {}_nG$, the set of roots of unity in k , is infinite.

(b) *The general case.* For every $s \in S$, there are an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \rightarrow U$ such that $G' = G_{S'}$ is diagonalizable, say $G' = D_{S'}(M)$. By 4.5.0 we may reduce to G' , and therefore suppose that G itself is diagonalizable. It is obtained by base change

$$S \longrightarrow \operatorname{Spec}(\mathbb{Z})$$

from the absolute group $H = D_{\mathbb{Z}}(M)$. Part (a) shows that, for every $s \in \operatorname{Spec}(\mathbb{Z})$, the family $({}_nH_s)_{n>0}$ is schematically dense in H_s . It remains only to apply 4.4.²⁵ $\square$

Corollary 4.8.

- (a) *Let $u, v : H \rightarrow G$ be two homomorphisms of S -preschemes in groups, with H of multiplicative type²⁶ and of finite type. Suppose that, for every integer $n > 0$, the restrictions of u and v to ${}_nH$ are equal. Then $u = v$.*
- (b) *Let H_1, H_2 be two subgroups of multiplicative type and of finite type of a prescheme in groups G . If ${}_nH_1 = {}_nH_2$ for every integer $n > 0$, then $H_1 = H_2$.*

Proof. Part (a) follows from part (b) by considering the subgroups H_1, H_2 of $H \times G$ that are the graphs of u, v . For part (b), put

$$H = H_1 \cap H_2.$$

This is a subgroup prescheme of H_i , for $i = 1, 2$, and we must show that it equals H_i . The hypothesis says precisely that H contains all the ${}_nH_i$. We are therefore reduced to the next corollary. $\square$

Corollary 4.9. *Let G be an S -group of multiplicative type and of finite type, and let H be a subgroup prescheme containing ${}_nG$ for every $n > 0$. Then $H = G$.*

Proof. By 4.7, it is enough to prove that H is closed, or equivalently that it equals G set-theoretically. This reduces us to the case in which S is the spectrum of a field. Every subgroup prescheme of G is then closed, by VI_B, 1.4.2. $\square$

Remark 4.10. Under the hypotheses of 4.7, let $m > 0$ be an integer with the following properties: for every $s \in S$, m is not a power of the characteristic of $\kappa(s)$; and if G_s

²⁵Editor's note: this result will be needed in X, 4.3 for a non-locally-Noetherian base, namely $S = \operatorname{Spec}(\widehat{A} \otimes_A \widehat{A})$, where $\widehat{A}$ is the completion of the Noetherian local ring A .

²⁶Editor's note: "diagonalizable" has been replaced by "of multiplicative type."

has type M , every prime divisor of the torsion subgroup of M divides m . The second condition is automatically satisfied if G is a torus. The proof shows that in Theorem 4.7 and Corollary 4.8 and Corollary 4.9 one may restrict to the subgroups

$$m^r G, \quad r > 0.$$

Moreover, if G is smooth over S , then for every $s \in S$ there are an open neighborhood U of s and an integer $m > 0$, prime to every residue characteristic on U , that satisfy the preceding conditions for G_U . For instance, take m to be the order of the torsion subgroup of the type of G at s (cf. 1.4.1(a) and 2.1(e)). After restriction to U , the groups $m^r G$ are then finite and étale over S , and their family is schematically dense. In certain applications—in particular those involving Theorem 3.2 and Theorem 3.2 bis (cf. XI, 6), though not those involving Theorem 3.6 and Theorem 3.6 bis—this observation allows one to avoid Theorem 3.1 and its use of Hochschild cohomology.

5. Central Homomorphisms of Groups of Multiplicative Type

Lemma 5.0.²⁷ *Let $(A, \mathfrak{m})$ be a Noetherian local ring, put $S = \operatorname{Spec}(A)$, and let H be a finite scheme over S , say $H = \operatorname{Spec}(B)$, where B is finite over A . Let K be a subscheme of H such that reduction modulo $\mathfrak{m}^{n+1}$ gives $K_n = H_n$ for every n . Then $K = H$.*

Proof. Let s be the closed point of S . First, K is a closed subscheme of H . Indeed, a priori it is a closed subscheme of an open subset U of H . But K , and hence U , contains the fiber

$$K_s = H_s.$$

Since $H \rightarrow S$ is finite, and therefore closed, the complement of U is empty; thus $U = H$. Hence K is defined by an ideal I of B . The hypothesis implies

$$I \subset \mathfrak{m}^n B \quad \text{for every } n.$$

Since B is a finite A -module, it is separated for the $\mathfrak{m}$ -adic topology. Therefore $I = 0$. $\square$

Theorem 5.1. *Let $u, v : H \rightarrow G$ be two homomorphisms of S -preschemes in groups, where H is of multiplicative type and of finite type. Suppose either that S is locally Noetherian or that G is of finite presentation over S . Let $s \in S$ satisfy $u_s = v_s$, and suppose that u_s is central. Then there is an open neighborhood U of s such that $u_U = v_U$.*

Proof. We distinguish the two cases.

(a) S locally Noetherian.²⁸ Let

$$K = \operatorname{Ker}(u, v)$$

be the inverse image of the diagonal of $G \times_S G$ under $(u, v) : H \rightarrow G \times_S G$. It is a subgroup prescheme of H . We seek an open U for which $K_U = H_U$. Because S is locally Noetherian and H is of finite type over S , the prescheme H is locally Noetherian. Hence the immersion $K \hookrightarrow H$ is of finite type (cf. EGA I, 6.3.5), so K is of finite type, and therefore of finite presentation, over S . By EGA IV₃, 8.8.2.4, it is enough to prove that

$$K_{S_0} = H_{S_0}, \quad S_0 = \operatorname{Spec}(\mathcal{O}_{S,s}).$$

²⁷Editor's note: this lemma has been made explicit because it is used several times below; it was implicit in the original.

²⁸Editor's note: the original has been expanded in what follows.

We may thus suppose that S is local with closed point s . Replacing the Noetherian local ring A by its completion $\hat{A}$, if desired, gives a faithfully flat quasi-compact base change $\hat{S} \rightarrow S$, and allows us to suppose that A is complete.²⁹

By 3.4, one has $u_n = v_n$ for every n , where the subscript denotes reduction modulo $\mathfrak{m}^{n+1}$, with $\mathfrak{m}$ the maximal ideal of A . For every $m > 0$, let

$${}_m u, {}_m v : {}_m H \longrightarrow G$$

be the homomorphisms induced by u, v . Then

$$({}_m u)_n = ({}_m v)_n \quad \text{for every } n.$$

The group ${}_m H$ is finite over S , by 2.2, so 5.0 gives ${}_m u = {}_m v$. This holds for every m , and therefore 4.7 gives $u = v$.

(b) *G of finite presentation.*³⁰ The group H is also of finite presentation over S . By EGA IV₃, 8.8.2, we may again suppose that S is local with closed point s , and it suffices to prove $u = v$. Let $f : S' \rightarrow S$ be a faithfully flat quasi-compact morphism. Write

$$f_H : H' \rightarrow H, \quad f_G : G' \rightarrow G, \quad u', v' : H' \rightarrow G'$$

for the resulting morphisms. If $u' = v'$, then

$$u \circ f_H = f_G \circ u' = f_G \circ v' = v \circ f_H,$$

and $u = v$ because f_H is an epimorphism. After a faithfully flat quasi-compact extension of the base, we may therefore suppose that H is diagonalizable:

$$H = D_S(M)$$

for a commutative group M of finite type.

As in the proof of 4.6, introduce the increasing filtered family of $\mathbb{Z}$ -subalgebras A_i of $A = \mathcal{O}_{S,s}$ that are of finite type, and put

$$S_i = \text{Spec}(A_i).$$

³⁰ For every i , the group $H = D_S(M)$ comes from

$$H_i = D_{S_i}(M).$$

Since G is of finite presentation over S , EGA IV₃, 8.8.2 (see also VI_B, 10.2 and 10.3) gives an index i , an S_i -group prescheme G_i of finite presentation, and homomorphisms

$$u_i, v_i : H_i \longrightarrow G_i$$

whose base changes are u, v . Let s_i be the image of s in S_i , and define

$$\rho_i : H_i \times_{S_i} G_i \longrightarrow G_i, \quad \rho_i(h, g) = u_i(h)g u_i(h)^{-1}.$$

Because u_s is central,

$$\rho_s = \rho_i \times_{\kappa(s_i)} \kappa(s)$$

²⁹Editor's note: this is because $K = H$ if $K_{\hat{S}} = H_{\hat{S}}$; cf. SGA 1, VIII, 5.4. The argument below does not use completeness of A .

³⁰Editor's note: the original has been expanded in what follows.

is the second projection. The extension $\kappa(s_i) \rightarrow \kappa(s)$ is faithfully flat, so ρ_i is also the second projection over s_i ; hence $(u_i)_{s_i}$ is central. Likewise $u_s = v_s$ implies

$$(u_i)_{s_i} = (v_i)_{s_i}.$$

Part (a), applied over S_i , now gives the conclusion. $\square$

Corollary 5.2. *Let $u : H \rightarrow G$ be a homomorphism of S -preschemes in groups, where H is of multiplicative type and of finite type, and suppose either that S is locally Noetherian or that G is of finite presentation over S . If $s \in S$ and u_s is the unit homomorphism, then there is an open neighborhood U of s on which u_U is the unit homomorphism.*

Corollary 5.3. *Let u, H, G be as in 5.2, and suppose in addition that G is separated over S . Let U be the set of points $s \in S$ for which u_s is the unit homomorphism. Then U is open and closed in S , and $u_U : H_U \rightarrow G_U$ is the unit homomorphism.*

Proof. Only closedness remains to be proved. Put $K = \text{Ker}(u)$. Since G is separated over S , K is a closed subprescheme of H , and U is the set of points $s \in S$ for which $K_s = H_s$. It follows readily that U is closed, for example by VIII, 6.4, since H is essentially free over S by VIII, 6.3. Alternatively, we may suppose that S is reduced and that U is dense in S , hence schematically dense. Then H_U is schematically dense in H because H is flat over S ,³¹ and u and the unit homomorphism, being equal on H_U , are equal on H . $\square$

Corollary 5.4. *Let S be a prescheme; let H, G be S -groups, with H of multiplicative type and of finite type and G separated and of finite presentation over S ; and let $\pi : S' \rightarrow S$ be a³² universal effective epimorphism—for example, a faithfully flat quasi-compact morphism or a morphism admitting a section (cf. IV, 1.12)—whose fibers are geometrically connected.³³*

Let $u' : H' \rightarrow G'$ be a central homomorphism of the S' -groups obtained from H, G by base change. Then there is a unique homomorphism $u : H \rightarrow G$ of S -groups such that $u \times_S S' = u'$. If S'/S admits a section g , then u is the morphism obtained from u' by the base change $g : S \rightarrow S'$.

Proof. Because π is a universal effective epimorphism, so is $H' \rightarrow H$; uniqueness follows as at the beginning of the proof of 5.1(b). If π has a section g , then

$$u' = \pi^* u \implies u = g^* \pi^* u = g^* u'.$$

For existence, IV, 2.3 reduces us to showing that the two homomorphisms

$$u_1'', u_2'' : H'' \longrightarrow G''$$

of S'' -groups obtained from u' by the two base changes

$$\text{pr}_1, \text{pr}_2 : S'' = S' \times_S S' \longrightarrow S'$$

are equal. Their pullbacks along the diagonal $S' \rightarrow S''$ are equal, since both are u' .³⁴ Because u_1'', u_2'' are central, we may apply 5.3 to

$$u_1''(u_2'')^{-1}.$$

³¹Editor's note: and of finite presentation over S ; cf. EGA IV₃, 11.10.5(ii) and 11.9.10(ii).

³²Editor's note: "covering morphism for the faithfully flat quasi-compact topology" has been replaced by "universal effective epimorphism," so that the result applies to the morphism $K \rightarrow S$ in 5.5; the beginning of the proof has been modified accordingly.

³³Editor's note: this holds, for example, when $S = \text{Spec}(k)$ and $S' = \text{Spec}(k')$, where k is a field and k'/k is radical; cf. X, Proposition 1.4.

³⁴Editor's note: the preceding sentence has been added.

There is therefore an open-and-closed subset U of S'' , containing the diagonal, over which u_1'' and u_2'' coincide. The fibers of S'/S are geometrically connected, hence so are those of S''/S , and in particular they are connected. Since U contains the diagonals of these fibers, it contains every fiber and therefore equals S'' . This proves existence. $\square$

Corollary 5.5. *Let S be a prescheme, let K be an S -group prescheme locally of finite type³⁵ and with connected fibers, and let H be an invariant subgroup of K , of multiplicative type and of finite type. Then H is a central subgroup of K .*

*Proof.*³⁶ First, $\pi : K \rightarrow S$ has geometrically connected fibers, because for a group scheme locally of finite type over a field, connectedness implies geometric connectedness (cf. VI_A, 2.4). Apply 5.4 with $G = H$ and $S' = K$ to the homomorphism of K -groups

$$u' : H_K \longrightarrow H_K, \quad (h, k) \longmapsto (khk^{-1}, k).$$

It is central because H is commutative. Its pullback along the unit section $\epsilon : S \rightarrow K$ is the identity homomorphism of H . [The source prints “of K ” here; the displayed endomorphism of H_K shows that H is intended.] Corollary 5.4 therefore implies that u' itself is the identity on H_K . Thus H is central in K . $\square$

Let us state the variants of the preceding results for central subgroups of multiplicative type. Arguing as above, and using 3.2 bis, gives:

Theorem 5.1 bis. *Let G be an S -group prescheme, and let H_1, H_2 be two subgroup preschemes of multiplicative type and of finite type, with $(H_1)_s$ central. Suppose either that S is locally Noetherian or that G is of finite presentation over S . For every $s \in S$ such that $(H_1)_s = (H_2)_s$, there is an open neighborhood U of s for which $H_1|_U = H_2|_U$.*

Corollary 5.3 bis. *Under the preceding hypotheses, suppose moreover that G is separated over S . Let U be the set of points $s \in S$ for which $(H_1)_s = (H_2)_s$. Then U is open and closed in S , and $H_1|_U = H_2|_U$.*

Corollary 5.4 bis. *Let S be a prescheme, let G be a group prescheme separated and of finite presentation over S , let $S' \rightarrow S$ be a covering morphism for the faithfully flat quasi-compact topology with geometrically connected fibers, and let H' be a subgroup of multiplicative type and of finite type of $G' = G \times_S S'$. Then there is a unique subgroup H of G such that $H' = H \times_S S'$. The subgroup H is of multiplicative type and of finite type over S , by 1.1 and 2.1(b). If $S' \rightarrow S$ admits a section g , then $H = g^*H'$.*

Theorem 5.6. *Let S be a prescheme and let $u : H \rightarrow G$ be a homomorphism of S -preschemes in groups, where H is of multiplicative type and of finite type and G is of finite presentation over S .*

- (a) *Suppose that G has connected fibers. If $s \in S$ and $u_s : H_s \rightarrow G_s$ is central, then there is an open neighborhood U of s such that the homomorphism $H|_U \rightarrow G|_U$ induced by u is central.*
- (b) *If $u_s : H_s \rightarrow G_s$ is central for every $s \in S$, then u is central.*

Proof. We first prove (a).³⁷ Arguing exactly as in the proof of 5.1(b), we may suppose first that S is local with closed point s , then that $H = D_S(M)$ is diagonalizable, and finally that there exist a finite-type $\mathbb{Z}$ -subalgebra A_i of $A = \mathcal{O}_{S,s}$ and a homomorphism

$$u_i : D_{S_i}(M) \longrightarrow G_i$$

³⁵Editor’s note: the hypothesis “locally of finite type” has been added to agree with VI_A, 2.4. In fact, over a field every connected group scheme is geometrically connected.

³⁶Editor’s note: the original has been expanded in what follows.

³⁷Editor’s note: the original has been expanded in the proof of (a).

whose base change is u and for which $(u_i)_{s_i}$ is central, where s_i is the image of s in S_i . We are thereby reduced to the case in which S is Noetherian local with closed point s , and we must prove that u is central.

Let $K = \text{Ker}(v, w)$, where

$$v, w : H \times_S G \rightrightarrows G$$

are defined by

$$v(h, g) = g, \quad w(h, g) = \text{int}(u(h)) \cdot g = u(h)g u(h)^{-1}.$$

Then K is a subgroup over G of

$$H_G = H \times_S G,$$

and we must prove that $K = H_G$. By 4.9, it is enough to show that K contains

$${}_m(H_G) = ({}_m H) \times_S G$$

for every $m > 0$. We are thus reduced to the case $H = {}_m H$, so that H is finite over S .

Let e be the unit element of the fiber G_s , and put

$$S^0 = \text{Spec}(\mathcal{O}_{G,e}), \quad S_n^0 = S^0 \times_S S_n, \quad S_n = \text{Spec}(A/\mathfrak{m}^{n+1}).$$

The scheme S^0 is Noetherian local because G is of finite presentation over the Noetherian scheme S . The base change

$$K_{S^0} = K \times_G S^0$$

is a subscheme of

$$H_{S^0} = H \times_S S^0,$$

and 3.9 gives

$$K_{S_n^0} = H_{S_n^0} \quad \text{for every } n.$$

³⁸ Since H_{S^0} is finite over S^0 , 5.0, applied to the Noetherian local ring $\mathcal{O}_{G,e}$, gives

$$K_{S^0} = H_{S^0}.$$

[The source prints the undefined local ring B here; the Noetherian local ring $\mathcal{O}_{G,e}$ just introduced is intended.]

On the other hand, $H \times_S G$ is Noetherian, being of finite presentation over the Noetherian scheme S . Thus the immersion

$$K \hookrightarrow H \times_S G$$

is of finite type (cf. EGA I, 6.3.5), and K is of finite presentation over G . The equality

$$K_{S^0} = H_G \times_G S^0$$

therefore implies, by EGA IV₃, 8.8.2.4, that there is an open neighborhood W of e in G such that

$$K \times_G W = H_G \times_G W = H \times_S W.$$

Hence K contains the open neighborhood

$$V = H \times_S W$$

³⁸Editor's note: the original has been expanded and corrected, replacing "the unit section of $(G_H)_n$ over H_n " by $H_{S_n^0}$.

of the unit section of $G_H = G \times_S H$ over H .

For every $t \in H$, the fiber G_t is an algebraic group over $\kappa(t)$, hence Cohen–Macaulay by VI_A, 1.1.1 and therefore has no embedded components. It is also connected, hence irreducible by VI_A, 2.4, so its generic point is its unique associated point. EGA IV₂, 3.1.8 therefore shows that V_t is schematically dense in G_t . By 4.3, V is schematically dense in G_H .

Moreover, since K is an H -subgroup of G_H , it induces on each fiber G_h a subgroup K_h . This subgroup contains an open neighborhood of the unit element; because G_h is connected, it follows that $K_h = G_h$. Thus $K = G_H$ set-theoretically. Finally, K is a closed subscheme of G_H containing the schematically dense subscheme V , and hence $K = G_H$. This proves (a).

Part (b) follows directly from 3.9: to verify that a subscheme K of $P = G \times_S H$ equals P , it is enough to verify, after every base change $S' \rightarrow S$ in which S' is the spectrum of an Artinian quotient of a local ring of S , that $K' = P'$. $\square$

6. Monomorphisms of Groups of Multiplicative Type, and Canonical Factorization of a Homomorphism from Such a Group

Lemma 6.1. *Let S be a quasi-compact prescheme, and let G be a commutative S -group prescheme, of finite presentation and quasi-finite over S . Then there is an integer $n > 0$ such that*

$$n \cdot \text{id}_G = 0, \quad \text{that is,} \quad G = {}_n G.$$

Proof. If S is the spectrum of a field k , then G is finite over k , and by VII_A 8.5 it is enough to take

$$n = \deg(G/k).$$

Suppose now that $S = \text{Spec}(A)$, where A is local Artinian. Let k be the residue field of A , put

$$G_0 = G \otimes_A k, \quad n_0 = \deg(G_0/k),$$

and distinguish two cases.

(a) *k has characteristic zero.* Then G_0 is separable over k , and hence G is unramified over S . The unit section of G is therefore an open immersion, so

$${}_n G = \text{Ker}(n \cdot \text{id}_G)$$

is an open subscheme of G . For it to equal G , it is consequently enough that it do so set-theoretically; thus one may take $n = n_0$.

(b) *k has characteristic $p > 0$.* Let $\mathfrak{m}$ be the maximal ideal of A , and put

$$S_r = \text{Spec}(A/\mathfrak{m}^{r+1}) \quad (r \geq 0).$$

Choose an integer m such that $\mathfrak{m}^{m+1} = 0$. We claim that one may take $n = p^m n_0$. Indeed, arguing by induction on m and putting $n' = p^{m-1} n_0$, we may suppose that $n' \cdot \text{id}_G$ induces the zero endomorphism on

$$G_{m-1} = G \times_S S_{m-1}.$$

³⁹ Let E be the set of endomorphisms u of G having this property, and let K be the kernel of the morphism of group functors

$$G \longrightarrow \prod_{S_{m-1}/S} G$$

(cf. III 0.1.2). Then

$$E = \operatorname{Hom}_{S\text{-gr}}(G, K);$$

in particular, the abelian group law on E is induced by that on K . By III 0.9, K is the S -group functor which, to every $g : T \rightarrow S$, assigns the k -vector space

$$\operatorname{Hom}_{\mathcal{O}_{T_0}}(g_0^*(\Omega_{G_0/S_0}^1), \mathfrak{m}^m \mathcal{O}_T).$$

Thus $pu = 0$ for every $u \in E$. It follows that

$$pn' \cdot \operatorname{id}_G = 0, \quad \text{that is,} \quad p^m n_0 \cdot \operatorname{id}_G = 0.$$

Suppose now that S is Noetherian; this is the case to which the general assertion reduces by the usual reduction to the Noetherian case.⁴⁰ It is enough to combine the foregoing with the following lemma.

Lemma 6.2. *Let X be a quasi-finite prescheme over a Noetherian prescheme S , and let $(X_i)_{i \in I}$ be an increasing filtered family of subpreschemes having the following property:*

$$\left\{ \begin{array}{l} \text{for every } s \in S \text{ and every } n \geq 0, \text{ putting} \\ S_{s,n} = \operatorname{Spec}(\mathcal{O}_{S,s}/\mathfrak{m}^{n+1}), \text{ there is an } i \in I \text{ such that} \\ X_i \times_S S_{s,n} = X \times_S S_{s,n}. \end{array} \right.$$

Then there is an $i \in I$ such that $X_i = X$.

Proof. Since S is Noetherian, there is a maximal open subset U on which

$$X|_U = X_i|_U$$

for all sufficiently large i . We shall prove that $U = S$. In other words, if $U \neq S$, we shall find an open subset U' strictly larger than U , and an i , such that

$$X|_{U'} = X_i|_{U'}.$$

Localizing at a maximal point s of $S - U$, we reduce to the case in which S is local with closed point s , and

$$U = S - \{s\}.$$

Indeed, put $S' = \operatorname{Spec}(\mathcal{O}_{S,s})$. If

$$X \times_S S' = X_i \times_S S'$$

for some i ,⁴¹ there is an open neighborhood V of s such that

$$X|_V = X_i|_V.$$

³⁹Editor's note: the original has been expanded in what follows, with the results of Exposé III that are used made explicit.

⁴⁰Editor's note: since S is quasi-compact, it is covered by finitely many affine open subsets, so one reduces to $S = \operatorname{Spec}(A)$. Since G is of finite presentation over S , EGA IV₃, 8.8.2 may then be applied.

⁴¹Editor's note: $X - \{s\}$ has been corrected above to $S - \{s\}$. Recall also that a quasi-finite morphism is assumed to be of finite type; cf. EGA II, 6.2.3, and EGA III₂, Err_{III} 20 for the definition of "locally quasi-finite." Thus here, since S is Noetherian, X is of finite presentation over S ; every immersion $X_i \hookrightarrow X$ is of finite type by EGA I, 6.3.5, so X_i is of finite presentation over S , and EGA IV₃, 8.8.2 applies.

Taking i large enough that $X_i|_U = X|_U$, we then have

$$X|_W = X_i|_W, \quad W = U \cup V.$$

For all sufficiently large i , the equality $X|_U = X_i|_U$ shows that X_i is a closed subscheme of X , defined by an ideal $\mathcal{J}^{(i)}$ whose support is contained in

$$X_s = X_0 = X \times_S S_0,$$

where

$$S_n = \operatorname{Spec}(A/\mathfrak{m}^{n+1}), \quad S = \operatorname{Spec}(A).$$

Since X_0 is quasi-finite over S_0 , it is a finite closed subset of the Noetherian prescheme X . Hence $\mathcal{J}^{(i)}$ is a module of finite length. There is therefore an integer $n \geq 0$ such that

$$\mathcal{J}^{(i)} \cap \mathfrak{m}^{n+1} \mathcal{O}_X = 0.$$

On the other hand, by the hypothesis of the lemma, after increasing i if necessary we may suppose that the image of $\mathcal{J}^{(i)}$ in

$$\mathcal{O}_{X_n} = \mathcal{O}_X / \mathfrak{m}^{n+1} \mathcal{O}_X$$

is zero. It follows that $\mathcal{J}^{(i)} = 0$, and therefore $X_i = X$. $\square$

Lemma 6.3. *Let S be a prescheme, let K be a group prescheme over S , locally of finite presentation, and let $s \in S$. Suppose that K_s is quasi-finite (respectively, unramified) over $\kappa(s)$ at the unit element. Then there is an open neighborhood U of s such that $K|_U$ is locally quasi-finite (respectively, unramified) over U .**

Proof. Let V be the set of points $x \in K$ having the following property: if t is the image of x in S , then K_t is quasi-finite (respectively, unramified) over $\kappa(t)$ at x ; that is, x is isolated in K_t (respectively, it is isolated and its local ring in K_t is a separable extension of $\kappa(t)$). The set V is open because K is locally of finite presentation over S .⁴² If ϵ denotes the unit section of K , then $\epsilon^{-1}(V)$ is open. By hypothesis it contains s , and is therefore an open neighborhood U of s . This U has the required property. Indeed, if $t \in U$, then K_t , being a group locally of finite type over $\kappa(t)$, is locally quasi-finite (respectively, unramified) over $\kappa(t)$ because it has that property at $\epsilon(t)$; cf. VI_B 1.3. $\square$

Combining 6.1 and 6.3, one obtains:

Theorem 6.4. *Let S be a prescheme, let H be an S -group of multiplicative type and of finite type, and let K be a closed group subscheme of H , of finite presentation over S . Let $s \in S$ be such that K_s is finite. Then there are an open neighborhood U of s and an integer $n > 0$ such that*

$$K|_U \subset {}_n H|_U.$$

In particular, since ${}_n H$ is finite over S , the group $K|_U$ is finite over U .

Nakayama's lemma now gives:

Corollary 6.5. *With the preceding notation, suppose that K_s is the unit group. Then there is an open neighborhood U of s such that $K|_U$ is the unit group.*

Corollary 6.6. *Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, where H is of multiplicative type and of finite type and G is separated over S . Suppose in addition*

*Cf. VI_B 2.5(ii) for a more general statement.

⁴²Editor's note: cf. EGA IV₃, 13.1.4 and EGA IV₄, 17.4.1. In addition, t , rather than s , has been used for the image of x .

that either S is locally Noetherian or G is locally of finite presentation over S .⁴³ If $s \in S$ and

$$u_s : H_s \longrightarrow G_s$$

is a monomorphism, then there is an open neighborhood U of s such that u induces a monomorphism

$$H|_U \longrightarrow G|_U.$$

Proof. Put $K = \text{Ker}(u)$. The hypothesis on u_s says that K_s is the unit group, and the desired conclusion says that $K|_U$ is the unit group for a suitable U . Since G is separated over S , the group K is a closed subgroup of H . If S is not assumed locally Noetherian, but G is assumed locally of finite presentation over S , then K is locally of finite presentation over S .⁴³ It is therefore of finite presentation over S , since it is separated over S (as H is) and quasi-compact over S (being closed in H , which is quasi-compact over S). Corollary 6.5 now applies. $\square$

Remark 6.6.1.⁴⁴ Under the hypotheses of 6.6, note that if, in addition, G is affine over S (respectively, of finite presentation over S), then

$$H|_U \longrightarrow G|_U$$

is even a closed immersion, as was observed in 2.5 (respectively, 2.6).

Corollary 6.7. Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, where H is of multiplicative type and of finite type. Suppose that either S is locally Noetherian or G is locally of finite presentation over S . If every homomorphism on fibers

$$u_s : H_s \longrightarrow G_s$$

is a monomorphism, then u is a monomorphism.

Proof. The argument is the same as in 6.6. Here the hypothesis that G be separated over S is unnecessary in order to ensure that $K = \text{Ker}(u)$ is closed in H , since the assumption that all the u_s are monomorphisms implies that K is set-theoretically reduced to the unit section. $\square$

Theorem 6.8. Let $u : H \rightarrow G$ be a homomorphism of S -group preschemes, where H is of multiplicative type and of finite type and G is separated over S . Suppose in addition that either S is locally Noetherian or G is of finite presentation over S . Then

$$K = \text{Ker}(u)$$

is a subgroup of H of multiplicative type and of finite type, and u factors as

$$H \xrightarrow{u'} H/K \xrightarrow{u''} G,$$

where H/K is of multiplicative type and of finite type, u' is the canonical homomorphism and is faithfully flat (and affine), and u'' is a monomorphism.

N.B. As noted in 6.6.1, the homomorphism u'' is a closed immersion if G is affine over S , or if G is of finite presentation over S .

⁴³Editor's note: it is enough to suppose that G is locally of finite type over S . By EGA IV₄, 1.4.3(v), this implies, since $H \rightarrow S$ is of finite presentation, that $H \rightarrow G$ is locally of finite presentation; the same is then true of $K \rightarrow S$, which is obtained from it by base change.

⁴⁴Editor's note: the number 6.6.1 has been added for use in later references.

Proof. It is enough to prove that K is of multiplicative type; the rest then follows from 2.7 and IV 5.2.6.

Suppose first that G is of finite presentation over S . Since this hypothesis is stable under base change, in proving that K is of multiplicative type we may reduce to the case in which H is diagonalizable:

$$H = D_S(M),$$

where M is a commutative group of finite type. Let $s \in S$. The group K_s is a closed subgroup of

$$H_s = D_{\kappa(s)}(M).$$

By 8.1,⁴⁵ it is therefore of the form $D_{\kappa(s)}(N)$, where N is a quotient of M . Put

$$K' = D_S(N).$$

Then K' is a subgroup of H of multiplicative type. Let

$$v' : K' \longrightarrow G$$

be the homomorphism induced by u .⁴⁶ By construction, v'_s is the unit homomorphism. Corollary 5.2 therefore gives an open neighborhood U of s such that

$$v'|_U : K'|_U \longrightarrow G|_U$$

is the unit homomorphism. Replacing S by U , we may suppose that v' itself is the unit homomorphism, and hence that u factors as

$$H \xrightarrow{u'} H/K' \xrightarrow{u''} G.$$

The homomorphism u''_s is obtained from u_s by factoring through

$$H_s \longrightarrow H_s / \text{Ker}(u_s),$$

and is therefore a monomorphism by IV 5.2.6. By 6.6, after restricting to a suitable open neighborhood U of s , the homomorphism

$$u''|_U : (H/K')|_U \longrightarrow G|_U$$

is a monomorphism. Thus, after this restriction,

$$\text{Ker}(u) = \text{Ker}(u') = K',$$

which proves that $\text{Ker}(u)$ is of multiplicative type.

The same proof applies if, instead of assuming G of finite presentation over S , one assumes S locally Noetherian, at least when H is diagonalizable. Without this additional hypothesis on H , one must show that there is a covering base change

$$S' \longrightarrow S,$$

with S' locally Noetherian, which trivializes H . This will be proved in the next exposé, X 4.6. □

⁴⁵Editor's note: the erroneous reference to IV 4.7.5 has been corrected. Note that Section 8 of the present exposé is independent of Sections 2 through 7.

⁴⁶Editor's note: "induced by G " has been corrected to "induced by u ."

7. Algebraicity of Formal Homomorphisms into an Affine Group

Theorem 7.1. *Let A be a Noetherian ring equipped with an ideal I for which A is separated and complete in the I -adic topology. Put*

$$S = \operatorname{Spec}(A), \quad S_n = \operatorname{Spec}(A/I^{n+1}),$$

and let H, G be S -group preschemes, with H of isotrivial multiplicative type and G affine. Then the canonical map

$$(x) \quad \theta : \operatorname{Hom}_{S\text{-gr}}(H, G) \longrightarrow \varprojlim_n \operatorname{Hom}_{S_n\text{-gr}}(H_n, G_n)$$

is bijective, where H_n, G_n are the S_n -groups obtained from H, G by the base change $S_n \rightarrow S$.

Proof. Suppose first that H is diagonalizable, hence of the form

$$H = \operatorname{Spec} A(M),$$

where $B = A(M)$ is the algebra of the commutative group M with coefficients in A . We may also write

$$G = \operatorname{Spec}(C),$$

where C is an A -algebra equipped with a diagonal map satisfying the usual axioms. Homomorphisms of S -groups $H \rightarrow G$ correspond bijectively to homomorphisms of A -algebras

$$\varphi : C \longrightarrow B$$

compatible with the diagonal maps; that is, such that, for every $f \in C$,

$$\Delta_H(\varphi(f)) = (\varphi \otimes \varphi)(\Delta_G(f)).$$

There is an analogous description of the homomorphisms of S_n -groups $H_n \rightarrow G_n$, given by homomorphisms of A_n -algebras

$$\varphi_n : C_n \longrightarrow B_n,$$

where

$$A_n = A/I^{n+1}, \quad B_n = B \otimes_A A_n, \quad C_n = C \otimes_A A_n.$$

[The source prints $C = C \otimes_A A_n$ in the last equality; C_n is intended.] Put

$$\widehat{B} = \varprojlim_n B_n, \quad \widehat{C} = \varprojlim_n C_n.$$

Then

$$\widehat{B} = \varprojlim_n A_n(M)$$

identifies with a submodule of the product A^M , namely the submodule of families $(a_m)_{m \in M}$ of elements of A that tend to zero for the I -adic topology along the filter of complements of finite subsets of M . In particular, the canonical homomorphism

$$B \longrightarrow \widehat{B}$$

is injective.⁴⁷ The projective system

$$\theta(\varphi) = (\varphi_n)_{n \geq 0}$$

defines a homomorphism

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B}$$

such that the following diagram is commutative:

$$\begin{array}{ccc} C & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ \widehat{C} & \xrightarrow{\widehat{\varphi}} & \widehat{B}. \end{array}$$

We prove that θ is surjective. Let $(\varphi_n)_{n \geq 0}$ be a projective system; we must show that it is obtained by reduction from some φ in the left-hand side of (x). A priori, the system (φ_n) defines a homomorphism on the completed algebras,

$$\widehat{\varphi} : \widehat{C} \longrightarrow \widehat{B}.$$

It is enough to prove that the composite

$$\Phi : C \longrightarrow \widehat{C} \xrightarrow{\widehat{\varphi}} \widehat{B}$$

sends C into B . Indeed, in that case it gives a homomorphism of A -algebras

$$\varphi : C \longrightarrow B$$

whose reductions are the φ_n . This homomorphism is immediately seen to be compatible with the diagonal maps, since the φ_n are so, and since

$$B \otimes_A B \longrightarrow \widehat{B \otimes_A B}$$

is injective, as follows from the same argument after replacing M by $M \times M$.

Passing to the completions, the diagonal homomorphism Δ_H defines

$$\widehat{\Delta}_H : \widehat{B} \longrightarrow \widehat{B \otimes_A B}.$$

Taking the projective limit of the analogous diagrams defined by the φ_n , one obtains the commutative diagram

$$\begin{array}{ccc} C & \xrightarrow{\Phi} & \widehat{B} \\ \Delta_G \downarrow & & \downarrow \widehat{\Delta}_H \\ C \otimes_A C & \xrightarrow{\Psi} & \widehat{B \otimes_A B}, \end{array}$$

where Ψ is the composite

$$C \otimes_A C \xrightarrow{\Phi \otimes \Phi} \widehat{B} \otimes_A \widehat{B} \longrightarrow \widehat{B \otimes_A B};$$

the last arrow is the evident canonical homomorphism. Consequently, for every $f \in C$, the element $\Phi(f) \in \widehat{B}$ has the property that its image under $\widehat{\Delta}_H$ is a “decomposable” element of $\widehat{B \otimes_A B}$, namely an element in the image of $\widehat{B} \otimes_A \widehat{B}$.

⁴⁷Editor’s note: the argument that follows has been expanded.

Let $(e_m)_{m \in M}$ be the canonical basis of $A(M)$, and let $(e_{m,m'})$ be that of

$$A(M \times M) = A(M) \otimes_A A(M).$$

Since

$$\Delta_H(e_m) = e_m \otimes e_m = e_{m,m} \quad (m \in M),$$

it remains only to apply the following lemma.

Lemma 7.2. *Let A be a Noetherian ring, let M be a set, and let $(a_{m,m'})$ be a family of elements of A , indexed by $M \times M$, satisfying:*

(i) $a_{m,m'} = 0$ if $m \neq m'$; that is, the support of the family is contained in the diagonal of $M \times M$;

(ii) one has

$$a_{m,m'} = \sum_i b_m^i c_{m'}^i,$$

where the b^i, c^i are finitely many elements of A^M ; in other words, $(a_{m,m'})$ belongs to the image of the canonical homomorphism

$$A^M \otimes_A A^M \longrightarrow A^{M \times M}.$$

Then the support of $(a_{m,m'})$ is finite.

Proof. By (i), the family $(a_{m,m'})$ is determined by the elements

$$a_m = a_{m,m}.$$

For $x = (x_n)_{n \in M} \in A(M)$, put

$$(u \cdot x)_m = \sum_{m'} a_{m,m'} x_{m'}.$$

Thus $(a_{m,m'})$ is viewed as the matrix of a homomorphism

$$u : A(M) \longrightarrow A^M.$$

Condition (i) gives simply

$$(u \cdot x)_m = a_m x_m.$$

On the other hand, condition (ii) gives

$$u \cdot x \in \sum_i A \cdot b^i.$$

Hence $u(A(M))$ lies in a finitely generated A -module. If $(e_m)_{m \in M}$ denotes the canonical basis of

$$A(M) \subset A^M,$$

then all the elements $a_m e_m$ lie in a finitely generated A -module. Since A is Noetherian, the module generated by these elements is itself finitely generated. Because the e_m are linearly independent, all but finitely many of the a_m are zero. This proves the lemma, and hence Theorem 7.1 when H is diagonalizable. $\square$

We now prove the general case of 7.1, in which H is only assumed isotrivial. There is then a finite étale surjective morphism

$$S' \longrightarrow S$$

such that

$$H' = H \times_S S'$$

is diagonalizable. We shall use only the fact that $S' \rightarrow S$ is finite and a covering for the faithfully flat quasi-compact topology, or simply for the canonical topology on **(Sch)**. Thus “étale” could be replaced by “flat.”

Put

$$S'' = S' \times_S S',$$

and similarly introduce

$$S'_n \quad \text{and} \quad S''_n = S'' \times_S S_n = S'_n \times_{S_n} S'_n,$$

together with $H', G', H'', G'', H'_n, G'_n, H''_n, G''_n$, obtained from H, G by the evident base changes. The groups H' and H'' are diagonalizable. Moreover, S' , and hence S'' , is affine. If

$$S' = \text{Spec}(A'), \quad S'' = \text{Spec}(A''),$$

then A' and A'' are separated and complete for the topologies defined by IA' and IA'' , respectively, because both are finite over A .

Since $S' \rightarrow S$ and $S'_n \rightarrow S_n$ are covering morphisms, we obtain a commutative diagram of maps of sets whose two rows are exact:

$$\begin{array}{ccccc} \text{Hom}_{S\text{-gr}}(H, G) & \longrightarrow & \text{Hom}_{S'\text{-gr}}(H', G') & \rightrightarrows & \text{Hom}_{S''\text{-gr}}(H'', G'') \\ \downarrow u & & \downarrow u' & & \downarrow u'' \\ \varprojlim_n \text{Hom}_{S_n\text{-gr}}(H_n, G_n) & \longrightarrow & \varprojlim_n \text{Hom}_{S'_n\text{-gr}}(H'_n, G'_n) & \rightrightarrows & \varprojlim_n \text{Hom}_{S''_n\text{-gr}}(H''_n, G''_n). \end{array}$$

The second row is exact because it is the projective limit of the exact diagrams associated with the various $S'_n \rightarrow S_n$. By the diagonalizable case already proved, u' and u'' are bijective. Thus u is bijective as well, completing the proof of 7.1. $\square$

Corollary 7.3. *Under the hypotheses of 7.1, suppose in addition that G is smooth over S , and let*

$$u_0 : H_0 \longrightarrow G_0$$

be a homomorphism of S_0 -groups. Then there is a homomorphism of S -groups

$$u : H \longrightarrow G$$

lifting u_0 . Any two such liftings u, u' are conjugate by an element $g \in G(S)$ whose reduction is the unit element of $G_0(S_0)$.

Proof. This follows by combining 3.6 and 7.1. To construct u , construct the u_n successively, as is possible by 3.6; Theorem 7.1 then shows that the system (u_n) comes from an S -homomorphism u . Given two liftings u, u' , construct successively elements g_n such that

$$g_0 = 1, \quad g_n \text{ is the reduction of } g_{n+1}, \quad u'_n = \text{int}(g_n)u_n.$$

Again, this is possible by 3.6. Since A is separated and complete, the g_n come from an element $g \in G(S)$. Finally, the injectivity in 7.1 gives

$$u' = \text{int}(g)u.$$

$\square$

Remark 7.4. Compare 7.1 with EGA III, 5.4.1, which implies that the statement of 7.1 remains true if, instead of assuming H of multiplicative type and G affine, one assumes that H is proper over S , and that G is separated and locally of finite type over S . A statement such as 7.1 without a properness hypothesis is rather exceptional, and should be understood here as one aspect of the great “rigidity” of the structure of a group of multiplicative type.

The analogous statement with

$$G = H = \mathbf{G}_a$$

is false in general. Indeed, take A of characteristic $p > 0$, and define the u_n , by reduction modulo I^{n+1} , from an additive formal power series

$$u(T) = \sum_{n \geq 0} a_n T^{p^n},$$

where the $a_n \in A$ tend to zero for the I -adic topology, but not for the discrete topology.

The statement of 7.1 also becomes false if the hypothesis that G be affine is omitted, even when $H = \mathbf{G}_m$. An example, with A a complete discrete valuation ring, is obtained from an elliptic curve over the fraction field K of A whose reduction, in the Néron–Kodaira theory, say, is the group $\mathbf{G}_m$ over the residue field k .⁴⁸ One then has a smooth commutative group scheme G over S , with special fiber $\mathbf{G}_{m,k}$. By 3.6 this permits the construction of a projective system of isomorphisms

$$u_n : H_n \xrightarrow{\sim} G_n, \quad H = \mathbf{G}_{m,A},$$

while the generic fiber of G is an abelian variety. Consequently, there is no homomorphism of S -groups $H \rightarrow G$ other than the zero homomorphism.

8. Subgroups, Quotient Groups, and Extensions of Groups of Multiplicative Type over a Field*

Scholium 8.0. ⁴⁹ — Let k be a field and let H be an affine k -group scheme. Write $k[H]$ for its affine algebra and

$$X(H) = \text{Hom}_{k\text{-gr}}(H, \mathbf{G}_{m,k})$$

for the group of characters of H . By the independence lemma for characters, $X(H)$ is a linearly independent subset of $k[H]$. Consequently, H is diagonalizable if and only if $X(H)$ spans $k[H]$ as a k -vector space.

Proposition 8.1. — Let G be a diagonalizable group scheme (respectively, a group scheme of multiplicative type) over a field k .⁵⁰ Then, for every subgroup scheme H of G , the schemes H and G/H are diagonalizable (respectively, of multiplicative type).

Definition 1.1 immediately reduces the assertion to the case without the parenthetical qualification. An easy passage to the limit, using VI_B 11.13 and VIII 3.1, reduces us to

⁴⁸Editor’s note: such an example could be made explicit.

*Added in July 1969. This afterthought section is independent of Sections 3 through 7.

⁴⁹Editor’s note: this scholium has been added.

⁵⁰Editor’s note: the words “over a field k ,” implicit in the original, have been added. In addition, “algebraic group” has been replaced by “group scheme,” since G is not assumed to be of finite type.

the case in which G is of finite type over k .⁵¹ By VI_B 11.16,⁵² one can find a finite family of nonzero elements

$$(8.1.1) \quad f_i = \sum_m a_{im} m, \quad a_{im} \in k,$$

of the affine ring $k(M)$ of $G = D_k(M)$, such that the points of H , with values in an arbitrary k -algebra k' , are precisely the points $g \in G(k')$ for which

$$(8.1.2) \quad \tau_g f_i = \lambda_i(g) f_i, \quad \lambda_i(g) \in k',$$

where τ_g denotes translation by g . But

$$(8.1.3) \quad \tau_g f_i = \sum_m a_{im} \chi_m(g) m,$$

where $\chi_m : G \rightarrow \mathbf{G}_{m,k}$ is the character associated with m . Thus (8.1.2) is equivalent to

$$(8.1.4) \quad \chi_m(g) = \chi_{m'}(g) \quad \text{if } m, m' \in Z_i,$$

where Z_i denotes the set of $m \in M$ for which $a_{im} \neq 0$. The same relation may be written

$$\chi_{m'-m}(g) = 1 \quad \text{if } m, m' \in Z_i.$$

Let N be the subgroup of M generated by all the elements $m' - m$, where i varies and $m, m' \in Z_i$. It follows from the definition of $D_k(M) \simeq G$ that H identifies with $D_k(M/N)$. Hence VIII 3.1 identifies G/H with $D_k(N)$. $\square$

Proposition 8.2. ⁵³ — *Let k be a field; let H and K be k -group schemes of multiplicative type and finite type; and let G be a k -group scheme fitting into an exact sequence*

$$1 \longrightarrow H \longrightarrow G \longrightarrow K \longrightarrow 1$$

(which implies that G is of finite type over k).

(i) *If G is commutative or K is connected, then G is of multiplicative type.*

(ii) *If K and H are diagonalizable and K is a torus, then G is diagonalizable.*

For the proof of (i), the reader is referred to XVII 7.1.1, of which 8.2(i) is a special case. The case of a field is treated in part (i) of the proof of XVII 7.1.1, without using the results of the later exposés.

For (ii), now suppose that K and H are diagonalizable:

$$K \simeq D_k(M) \quad \text{and} \quad H \simeq D_k(N),$$

and suppose that G is of multiplicative type, as follows from (i) when K is a torus. By X 1.4, G is isotrivial over k ; that is, there is a finite separable extension k'/k such that

$$G' = G \times_k k'$$

⁵¹Editor's note: this "passage to the limit" should be spelled out. It uses 8.0 and the equality

$$G/H = \varprojlim_i G_i/H_i$$

(cf. VI_B, proof of 11.17). To see that $H = \varprojlim_i (H \cap G_i)$, does one use the fact that H is closed in G ? Using this fact, one can give a direct proof.

⁵²Editor's note: VI_B 11.16 should doubtless be rewritten in the usual, more agreeable form. In particular, a single f_i suffices below.

⁵³Editor's note: both the statement and its proof have been expanded.

is diagonalizable. Hence $G' = D_{k'}(E)$ for a commutative group E , and there is an exact sequence

$$(1) \quad 0 \longrightarrow D_{k'}(N) \longrightarrow D_{k'}(E) \longrightarrow D_{k'}(M) \longrightarrow 0.$$

By VIII 3.1 and 3.2, M is therefore a subgroup of E , and one has an exact sequence

$$(2) \quad 0 \longrightarrow M \longrightarrow E \longrightarrow N \longrightarrow 0.$$

For a given extension E_0 of N by M , consider the diagonalizable k -group $G_0 = D_k(E_0)$. The k -group functor A of automorphisms of the extension

$$(3) \quad 1 \longrightarrow H \longrightarrow G_0 \longrightarrow K \longrightarrow 1,$$

that is, the group subfunctor of $\text{Aut}_{k\text{-gr}}(G_0)$ whose points on a k -prescheme T are the

$$\phi \in \text{Aut}_{T\text{-gr}}(G_T)$$

inducing the identity on H_T and K_T , identifies with $\text{Hom}_{k\text{-gr}}(K, H)$. By VIII 1.5, this is the constant k -group with value

$$L = \text{Hom}_{\text{gr}}(N, M).$$

It follows that the extensions G of K by H which become isomorphic to (3) over a separable closure k_s of k are classified in the same way as the k -torsors, for the étale topology, under the constant group L_k . Put $\Gamma = \text{Gal}(k_s/k)$. Since L is a trivial Γ -module, these torsors are classified by

$$H_{\text{ét}}^1(k, L_k) = H^1(\Gamma, L) = \text{Hom}_{\text{gr. top.}}(\Gamma, L).$$

If K is moreover a torus, then M is torsion-free, and hence so is L . Therefore

$$\text{Hom}_{\text{gr. top.}}(\Gamma, L) = 0,$$

and every extension of K by H is already diagonalizable.

Remark 8.3. — *As already noted in the proof of 8.2, the first assertion is stated and proved over an arbitrary base in XVII 7.1.1. The second assertion generalizes, with essentially the same proof, to the case of an integral normal base scheme—or, more generally, a geometrically unibranch one—by using X 5.13 below.*

*Finally, 8.1 may also be regarded as a corollary of the distinctly less trivial result 6.8, which is likewise valid over an arbitrary base scheme.*⁵⁴

Bibliography

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[BEns] N. Bourbaki, *Théorie des ensembles*, Chapters I–IV, Hermann, 1970.

⁵⁴Editor's note: the proof of 6.8 nevertheless uses 8.1.

⁵⁵Editor's note: additional references cited in this exposé.